# Savings and Migration in a Warming World*

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**Abstract**

This paper introduces a global model of climate change where agents can adapt through precautionary savings and migration. Analytical results show that pecuniary migration costs determine how migration will serve as adaptation to climate change and in turn alter the global population distribution. Adaptation through migration is income-dependent; while wealthy agents are able to pay international migration costs in order to take advantage of locations forecast to gain under climate change, poor and low-wealth agents are unable to move and can only partially insure against adverse shocks through savings. In the calibrated model, global warming under RCP 4.5 induces additional migration that leads to a 12 percentage point change in the global population distribution by the end of the 21st century. The welfare effects from climate change are highly unequal; the world's poorest decile experience losses equivalent to an 8.2 percent decline in lifetime consumption, while the wealthiest decile experience gains of 13.6 percent on a consumption-equivalent basis.

**Keywords:** Climate change, international migration, incomplete markets

**JEL Classification:** D52, F22, J61, Q54

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# 1 Introduction

Climate change due to anthropogenic greenhouse gas (GHG) emissions is forecast to substantially alter the spatial distribution of people and economic activity over the coming century (Lee et al. 2023). A broad literature has established that climate change affects output and investment (Yang 2023; Nath, Ramey, and Klenow 2023) as well as where people live (Hoffmann et al. 2020; Liu, Shamdasani, and Taraz 2023). The resulting shift in populations towards less-affected places will be governed by how adaptation—individuals' behavioral adjustments taking future warming as given—will manifest through relocation of people across space and resources over time (Desmet and Rossi-Hansberg 2024). The amount each adaptation channel, precautionary savings and migration, will be put in use across the globe depends on the local efficacy of each channel as well as the extent to which either is accessible. To date, the climate macroeconomics literature incorporating multiple regions has studied these channels in isolation.

This paper focuses on how the global population will adapt to the spatially-differentiated effects of climate change given the ability to migrate towards better-off locations and partially insure through precautionary savings. The neoclassical and spatial modeling paradigms in the multi-region climate macroeconomics literature have focused on how either local capital accumulation or migration will, in isolation, allow people to respond to climate change.[1] The neoclassical integrated assessment model (IAM) paradigm views adaptation to climate change through the lens of a consumption-savings problem, where the differentiated effects of climate change across space affect the allocation of local resources over time (Krusell and Smith 2022). While this approach is useful for characterizing how climate change affects adaptation through local capital accumulation, it forgoes modeling migration decisions.[2] The spatial integrated assessment model (SIAM) paradigm draws on the economic geography literature, abstracting from intertemporal savings and capital accumulation in favor of characterizing how migration, trade, and local sectoral compositions shape adaptation to climate change (Cruz and Rossi-Hansberg 2023; Cruz 2023). This paper asks how income and wealth determine movement: in a given location, who will use migration as a channel for adapting to climate change?

Towards integrating these two paradigms, I present a macroeconomic model of global climate change that retains the intertemporal choices of the neoclassical formulation while

[1] I give an explicit taxonomy in Section 1.2.

[2] A notable exception to this is Alsina-Pujols (2026) who examines how climate change will affect migration in the form of international refugee flows between the global North and South. Alsina-Pujols (2026) shows that Northern countries have in the past displayed limited willingness to accept refugees from the Global South and tend to experience political backlash in the aftermath. This channel increases Northern countries' willingness to pay for climate change mitigation as it also decreases Southern emigration.

allowing for agents to adapt to climate change through migrating across space. The central mechanism that differentiates my model from prior work is that explicit financial costs of international migration can preclude movement entirely for the poorest individuals. While migration in the calibrated model would be beneficial for many households in terms of gains in real income, credit constraints and prohibitively high costs may prevent them from being able to do so. This aspect of the model reflects a large body of literature showing international migration is increasing in income at low income levels where liquidity constraints play a critical role in preventing the poorest agents from moving (Cai et al. 2016; Bazzi 2017; Cai 2020; Barbosa-Alves and Braulio 2023).

The paper begins with a two-period model where agents can migrate and save in order to adapt to a changing climate. Limiting cases nest stylized versions of the neoclassical and economic geography models of global warming, namely when either migration costs are prohibitively high or the gross returns to savings are zero. Results diverge from those in traditional spatial models as consumption-based migration costs render places that are especially expensive to access infeasible for many initial locations. When migration costs are pecuniary, income determines the ability to migrate regardless of associated changes in utility or future gains in real incomes. In partial equilibrium, comparative statics align with intuition: emigration declines with migration costs, increases with local climate damages, and may be precluded entirely by liquidity constraints. The addition of the savings channel to the conventional spatial framework leads to all regions retaining a nonzero share of agents. This emergence of "always-stayers" under savings occurs regardless of the size of local losses from climate change; unless losses from a decision are infinite, the class of models that use unbounded taste shocks for tractability will always have agents for whom it's optimal to stay in place when they can guarantee some future income. In general equilibrium, the final distribution of modeled agents depends on the relationship between migration costs and future productivity levels across space. Identifying how each channel will be used for adaptation requires fully characterizing the distribution of migration costs, baseline productivity levels, and effects of climate change.

The full model takes these ideas to a calibrated infinite-horizon incomplete markets model where a continuum of agents is divided across countries around the globe. Each period, agents solve a traditional consumption-savings problem in addition to a discrete decision over whether to migrate internationally and where to move. Modeled countries differ in fundamental productivity levels, migration costs, and initial temperature levels. Agents in each region face idiosyncratic labor supply risk, leading to precautionary savings (as in Aiyagari 1994) and induces a rich cross-sectional distribution of wealth within and across countries. Unlike in other multiregion frameworks (cf. Cruz and Rossi-Hansberg 2023;

Krusell and Smith 2022; Hassler, Krusell, and Olovsson 2026), migration costs enter the model as financial costs rather than pure disutility. Whether it is visa fees, transportation costs alone, or "golden passports", this aspect reflects the reality that international migration is rarely free of charge (Recchi et al. 2021; Surak 2021). Real costs of movement interact with borrowing constraints so that poor agents must self-finance international migration. Pecuniary migration costs also lead idiosyncratic income shocks to affect migration decisions; while the relative wage of outside locations determines their value as insurance against future climate change, negative income shocks can (depending on local migration costs) preclude leaving (Bazzi 2017; Cai 2020). The combination of these features yields an environment where climate change affects local income levels as well as migration decisions differently for poor and rich agents in each country.

I calibrate an infinite-horizon version of the model with 154 countries to match the global population distribution between 2010-2014. Calibrated values for existing barriers to movement between countries—the real costs of international migration that minimize the distance between data and modeled moments—are substantially larger than annual wages for a qualitatively important share of agents. When translated to real world terms, these costs are in the thousands of U.S. dollars, values in line with estimates of barriers to movement from the empirical migration literature (Clemens, Montenegro, and Pritchett 2019). Migration costs of the magnitude suggested by the calibration would be prohibitively high for low-wealth agents in practice in the absence of the ability to borrow heavily against future income.

I use the calibrated model to perform counterfactual steady-state and transition path experiments that forecast how climate change in the 21$^{st}$ century will affect the global population distribution. The projected aggregate shocks from climate change driving the model are based on two benchmark scenarios for global GHG emissions, Representative Concentration Pathways (RCP) 4.5 and 8.5. I use grid-cell level temperature projections from the Coupled Model Intercomparison Project Phase 6 (CMIP6 – Eyring et al. 2016) to map each RCP into time series for annual changes in country-level average temperatures relative to initial conditions in the modeled 2010-2014 period. I combine these country-specific temperature forecasts with base-temperature specific damages functions estimated by Nath, Ramey, and Klenow (2023), henceforth NRK, to create forecasts for how climate change will affect productivity growth through 2100. These two inputs combine to yield RCP-specific changes in annual productivity growth for the entire transition period, which drive how agents behave as the world moves towards a new (warmer) equilibrium.

## 1.1 Results

The two-period model gives analytical results on how intertemporal substitution affects migration's role in adaptation. Savings allow agents to finance future consumption even when local labor income falls to zero, so idiosyncratic preferences lead every location to retain a positive share of its initial population regardless of local damages. At the same time, unlike when migration induces disutility alone, migration costs paid in consumption goods can make some destinations infeasible. Tighter borrowing limits can further restrict agents' ability to adapt through migration, as they become unable to finance migration even when real income gaps are large. Higher migration costs reduce emigration and larger local climate damages reduce the share of agents choosing the affected location. Higher returns to savings also increase the share of stayers in locations with below-average future productivity, although the effect is ambiguous elsewhere. These results motivate jointly modeling savings and migration: the relative importance of each adaptation channel depends on the distribution of productivity, climate damages, and barriers to movement across locations.

The calibrated model shows the global gradient of productivity changes induced by climate change under RCP 4.5 substantially shifts the new steady-state distribution of people across the world. Twelve percent of the world population relocates (on net) to different countries after the cumulative effects of warming through 2100 are fully realized. In contrast to prevailing narratives suggesting that large waves of emigration will take place in the Global South (Vince 2022), migration is most concentrated among rich countries. Forecast changes in populations in Sub-Saharan Africa, a region where baseline productivities are relatively low and the impacts of climate change are forecast to be very negative, are all lower than 0.1 percent of the global population. This is not a result of low overall rates of migration; many agents do migrate each period and the poorest agents are those who would otherwise gain the most. Instead, the high cost of international migration and limits on borrowing preclude the poorest agents from migrating even when they know they will face large declines in future real income. The largest population declines in the baseline model occur instead in the United States, Australia, Japan, and Western Europe. These declines are not due to the direct local effects of climate change on productivity in these countries, but rather because rich agents there have sufficient income vis-à-vis migration costs to take advantage of colder locations forecast to experience future productivity gains.

The dynamic experiment uses transition path methods to simulate the full sequence of counterfactual dynamics under RCP 4.5 over the next 200 years. Almost 6 percent of the global population (approximately 400 million people) migrate internationally on net in response to climate change within the 21$^{st}$ century. Like in other models (Bilal and Rossi-Hansberg 2023), anticipation effects prove important for determining population dynamics

when agents have rational expectations over the damages from climate change. On aggregate, the welfare losses from climate change are almost zero; adaptation through migration proves to be effective insurance (and indeed can provide gains) for agents who can move to cooler countries that become more productive. However, this small aggregate effect masks a highly unequal distribution of gains and losses at the individual level. A warmer world leads to welfare losses as high as 8.5 percent for poor agents in low-income countries, as these agents are unable to afford to migrate and are in places that are worst-hit by climate change. In contrast, rich agents who are not constrained can freely migrate to countries where productivity rises and experience welfare gains as large as 4 percent. On the whole, results from the transition path add to the growing body of research that suggests the welfare effects of climate change will be highly unequal by geography as well as across income groups within a given country.

The divergence in behavior between poor and rich agents becomes extreme when the counterfactuals are recalculated under the larger productivity changes due to RCP 8.5 or when in addition to the warming under RCP 4.5, migration costs are doubled. Population changes under RCP 8.5 (roughly 4.2 °C warming and 17 percentage points of productivity growth loss globally) are qualitatively similar in terms of where populations rise and fall relative to the benchmark case, albeit much more extreme in magnitude. Over 20 percent of the global population migrates internationally, with increases in population levels most concentrated in today's coldest countries that are forecast to experience the largest productivity gains from climate change. The model is sensitive enough to migration costs that the distribution of agents across space becomes close to a degenerate one under the scenario where, in addition to the warming under RCP 4.5 migration costs are doubled. Modeled agents fall into one of two equilibrium behaviors: poor agents around the equator become stuck almost entirely in their countries or regions of origin, while nearly all rich agents make it to one of the countries where real wages rise due to warming and then stay put because emigration due to idiosyncratic shocks rarely outweighs the costs of moving after they are doubled.

The key insight from the simple model is that the value of migration as adaptation will depend on the *financial* costs of migration in addition to any utility costs. When existing spatial approaches are modified to account for the nonlinearities that emerge when migration is not free of pecuniary costs, forecast patterns of adaptation to climate change may diverge substantially from what would otherwise be predicted by the effects on the global productivity gradient alone. This feature of the model illustrates the crucial role of how barriers to migration may lead to nonlinearities in the realized migration patterns over the coming century. As shown in Section 6.3, the world will look substantially different at

the end of the 21st century if climate change is accompanied by an increase in the pecuniary costs of migration. The resulting uneven uptake of migration across income and wealth levels drives the results here to diverge from those of existing SIAM frameworks examining climate change (Cruz and Rossi-Hansberg 2023; Cruz 2023). The ability to pay whatever the costs of international migration are in the coming century will play a non-negligible role in shaping what the world looks like as it warms and pressures on emigration build, and will distinguish between who we will observe moving as opposed to who would move if it were feasible.

## 1.2 Related Literature

**Neoclassical models:** Neoclassical models of climate change with multiple regions (e.g., Nordhaus and Yang 1996; Nordhaus 2010; Krusell and Smith 2022; Yang 2023) typically build on William Nordhaus' Dynamic Integrated Climate Economy (DICE) model framework (Nordhaus 1992). These models augment the Ramsey model of optimal savings to account for how greenhouse gas emissions from production, mitigation efforts, and abatement shape welfare and climate globally (Barrage and Nordhaus 2024). Representative agents across a set of (potentially many) regions solve consumption savings problems under autarky while facing a global externality in the form of climate change due to emissions from energy use. The optimal response for a policymaker in the decentralized setting is to adjust energy use in each region to lower emissions through the use of Pigouvian taxes. The current version of DICE that allows for multiple regions (Yang 2023) and the state-of-the-art multi-region neoclassical IAMs in terms of granularity (Krusell and Smith 2022; Hassler, Krusell, and Olovsson 2026) serve as preeminent guides for "place-based" policy in a world where space is relevant only in determining how climate change affects productivity and emissions. In a world where capital and labor cannot move between locations, the neoclassical paradigm solves for location-specific first-best policies. Counterfactuals in these models are then the distributional and global gains from imposing Pigouvian taxes relative to a laissez-faire baseline.

**Economic geography models:** The alternative leading approach uses tools from the economic geography literature, namely dynamic spatial general equilibrium model. The economic geography framework offers tractability, high spatial resolution, and a substantially broader choice set for modeled agents in exchange for abstracting from some aspects of intertemporal choice. Desmet and Rossi-Hansberg (2015) is (to my knowledge) the first of these models – a two sector IAM with costless mobility for labor within abstract borders – that examines the heterogeneous effects of climate change on migration, trade, and welfare.

State-of-the-art models building on this work (e.g., Nath (2020), Conte et al. (2021), Rudik et al. (2021), Burzyński et al. (2022), Cruz (2023), and Cruz and Rossi-Hansberg (2023)) incorporate more realistic geographies as well as local variation in fundamental amenities, productivity, and the frictions associated with the movement of goods and agents.[3] A drawback to this approach is that the functional forms for utility and investment are restrictive: bilateral migration is always feasible between any two locations as migration costs cannot bind, mobility decisions often collapse to static (rather than forward-looking) choices, and modeled workers typically live hand-to-mouth (i.e., are unable to shift consumption intertemporally). Agents' only mechanism in these models for insuring their future consumption is to move to a location with higher productivity levels.

**The (nascent) intersection:** This paper is intended to serve as a first step in bridging the gap between the two workhorse model paradigms. The closest work in the dynamic spatial general equilibrium literature considering climate change is Bilal and Rossi-Hansberg (2023) in which they form a county-level model of the United States that examines how climate change affects local capital accumulation while allowing workers to move between locations. In their model, the anticipated effects of climate change can shape current capital accumulation (and in turn, current wages and migration decisions). However, workers (the agents who migrate) remain hand-to-mouth agents who make decisions solely over mobility; while capital dynamics affect expectations, workers themselves are unable to alter their consumption/savings patterns as a channel of adaptation to climate change. Other work by Hassler, Krusell, and Olovsson (2026) adopts a tractable framework with frictionless migration within borders (which responds to shifts in productivity) but does not allow for migration internationally.

My paper also has similar aspects to that of Barbosa-Alves and Braulio (2023), who form a partial-equilibrium model of emigration from rural areas in Guatemala to the United States in response to climate shocks. Like my paper, theirs combines real migration costs and incomplete markets and shows that the value of migration as insurance against climate shocks is sensitive to household income levels. This paper abstracts from incorporating feedback between economic activity and climate change that would close my model as a traditional IAM. Unlike Krusell and Smith (2022) and Cruz and Rossi-Hansberg (2023), I assume a fixed pathway for global temperatures based on benchmark forecasts in the literature and abstract entirely from the endogenous effects of economic activity on climate change. I also abstract from agglomeration and crowding externalities present in other spatial models that

[3]The economic geography models exhibit an exceptional degree of richness. Attractive features include endogenous natality, trade, and migration on the agents' side along with new technology adoption and increasing marginal extraction costs faced by firms in the emitting sectors (Cruz and Rossi-Hansberg 2023).

are key determinants of location choice within countries (Desmet, Nagy, and Rossi-Hansberg 2018) and would provide additional degrees of freedom to discipline some of the results I present here. Further work integrating the richness of spatial models with the intertemporal substitution channel for adaptation that forms the basis of modern macroeconomics remains the frontier and a natural avenue for future research.

# 2 A Two-Period Warming World

This section presents a two-period version of a dynamic model of savings and migration as adaptation. The model highlights the importance of savings and mobility in a world where climate change shifts the distribution of local productivity levels over time. In the first period, agents receive their endowments and decide whether to migrate as well as how much to lend and borrow on the capital market. In the second period, the effects of climate change are realized. Agents receive factor income based on their savings and the prevailing local wages in each location. When returns to capital are exogenous, extreme cases of the model nest analogues of neoclassical and economic geography models. Comparative statics follow: emigration shares are increasing in local warming and declining in migration costs and the stringency of borrowing. Critically, when migration requires an up-front cost paid with real consumption goods, constraints on borrowing can prevent modeled agents from moving to areas less-affected by climate change. This result is contextualized in the simple setting and motivates the inclusion of each channel for adaptation in the fully calibrated model.

## 2.1 Model Environment

The world consists of $N$ of islands indexed by $i \in \{1, \ldots, N\}$. It lasts for two time periods, indexed 1 and 2. Climate change is realized in period 2, which causes each island to experience damages to local productivity levels. Each island is populated by a unit mass of *ex-ante* identical agents. Each atomistic agent has identical utility over consumption on each day and is endowed with a single unit of labor they supply inelastically. Let the function $L_t(i)$ denote the density of agents on island $i$ at a given point in time.

Agents face two decisions in the first period: whether to move to a different island, and how much of their first period income to save. Moving from location $i$ in the first period to location $j$ imposes two costs. First, they face a utility cost of $U > 0$ if they do not stay on island $i$. Second, they face a pecuniary cost, requiring them to forgo $m(i, j)$ units of first-day consumption. I assume this cost is non-negative, continuously differentiable when $i \neq j$, and

zero for agents that do not move ($m(i,i) = 0, \forall i$). The cost may be asymmetric such that $m(i,j)$ is not necessarily equal to $m(j,i)$. The limiting case, $m(i,j) = \infty, \forall i \neq j$ where no agents move, represents the most stylized version of the neoclassical model where pecuniary costs are functionally infinite.

Agents on all islands additionally have access to a global capital market that allows them to borrow or lend at the risk free rate $r$. Assume initially this market consists of a large enough number of borrowers and lenders such that the agents' collective savings decisions do not affect the interest rate. Saving (borrowing) $S$ of the consumption good in the first period allows them to receive (pay) $(1+r)S$ in the second period. When borrowing and savings are prohibited, the model resembles an economic geography framework.

Agents' income on the first day is a uniform endowment $y$ across all islands. On the first day, an agent on island $i$ faces a budget constraint over savings $S$, first-period consumption $c_1$, and the costs of migration $m$

$$y = S(i,j) + m(i,j) + c_1$$

where arguments $(i,j)$ denote the agent's savings and pecuniary migration costs conditional on starting in island $i$ and choosing to move to island $j$. First period borrowing is constrained by the natural borrowing limit (NBL) determined by second-period income:

$$S(i,j) \geq -\underbrace{w_2(j)(1+r)^{-1}}_{\text{natural borrowing limit (NBL)}} \tag{1}$$

as consumption cannot be negative and default is not l agents can't borrow beyond

In the second period, after agents have moved, each island experiences an idiosyncratic degree of damages from climate change. For simplicity, I assume the economic effects take the form of a shock to labor-augmenting productivity in the second period $A_2(i)$ that is given by a multiplicative exponential function:

$$A_2(i) = \exp\left(\frac{-\theta(i)}{1-\alpha}\right)$$

where climate damages $\theta(i)$ parameterize the extent to which climate change affects output in each location and $\alpha \in (0,1)$.[4] Agents have full information over the initial distribution of the population as well as the level of flooding that will occur on each island prior to production on the second day leaving no aggregate uncertainty.

[4]Note these effects can, in principle, be positive or negative. I implicitly assume all $\theta(i)$ are elements of the set $\{\bar{\mathbb{R}} \setminus -\infty\}$ such that wages are finite for all locations in the second period.

Production in the second period is facilitated by a representative firm on each island. All firms have access to identical Cobb-Douglas technology that uses capital $K$ and effective labor $AN$ to produce the consumption good:

$$Y_2(i) = K_2(i)^{\alpha}(A_2(i)N_2(i))^{1-\alpha}$$

Each representative firm may rent an unrestricted level of capital on the global market at rate $r$. Labor markets operate in autarky; the firm on each island may only hire labor from residents who reside there. As labor is supplied inelastically, in equilibrium demand for labor on each island is equal to the number of agents residing there. Equilibrium wages on each island on the second day are

$$w_2(i) = A_2(i)(1-\alpha)\left(\frac{\alpha}{r}\right)^{\frac{\alpha}{1-\alpha}}$$

Finally, I assume each atomistic agent in a given location $i$ draws an idiosyncratic set of additive taste shocks associated with each one of her potential next-period locations, $\{\lambda(j)\}_{j\in\{1,\ldots,N\}}$, prior to migration decisions. The shocks are independent draws from a Type-1 extreme value distribution with variance $\pi^2/6$ and scale parameter one recentered to be mean zero, lending a convenient closed form for the resulting policy functions.[5] Let $W(i,j)$ denote the conditional value function (defined as the return net of taste shocks) of moving for an agent who begins on island $i$ and moves to island $j$:

$$\begin{aligned} W(i,j) = \max_{S(i,j)} \Big\{ & (1-\beta)\ln\Big(c_1(i)\Big) \; - \; U\,\mathbb{I}\{j \neq i\} \; + \; \beta \; \ln\Big(c_2(j)\Big) \Big\} \\ \text{s.t.} \quad & c_1(i) = y - S(i,j) - m(i,j) \\ & c_2(j) = w_2(j) + (1+r)S(i,j) \end{aligned} \tag{2}$$

the expression in (2) is the maximized value of utility from consumption across each period for an agent who begins at location $i$ and migrates to location $j$ net of her idiosyncratic taste shocks. The discount factor $\beta \in (0,1)$ determines agents' relative weights on current and future consumption. Each agent who starts in location $i$ can express her two-period problem, the choice over her second-period location $j$, as a function of her taste shock draws and the conditional desirability of islands:

[5] This choice of functional form for preference shocks is ubiquitous in the discrete choice literature (Train 2009) and substantially aids tractability (Iskhakov et al. 2017) when taken to the calibrated model. It is not necessary for the quantitative results presented in Section 5, but it is central to the example here illustrating how pecuniary costs qualitatively effect the implications of spatial models.

$$V(i) = \max_{j \in \{1,\ldots,N\}} \left\{ \lambda(j) + W(i,j) \right\} \tag{3}$$

The probability an agent at location $i$ chooses location $j$ as her second period destination induced by the problem in (3) is given by a closed form function of exponentiated conditional values across all destinations. The probability of choosing destination $j$ given origin $i$ is:[6]

$$\pi(i,j) = \frac{e^{W(i,j)}}{\sum_{k=1}^{N} e^{W(i,k)}} \tag{4}$$

We can now define an equilibrium for this economy:

**Definition 1.** A spatial competitive equilibrium given damages $\{\theta(i)\}_{i=1}^{N}$ consists of savings policies $S(i,j)$, migration probabilities $\pi(i,j)$, prices $w_2(j)$ and $r$, and second-period allocations for each representative firm $(N_2(j), K_2(j))$ such that:

1. Taking prices as given, savings policies $S(i,j)$ and migration levels $\pi(i,j)$ maximize consumers' conditional problem for all location pairs $(i,j)$

2. Given prices and damages, firms' second-period decisions over hiring and capital rental $(N_2(j), K_2(j))$ are optimal in all locations

3. Migration levels $\pi(i,j)$ lead to second-period populations that clear labor markets

$$\sum_{i=1}^{N} \pi(i,j) = N_2(j), \qquad j = 1 \ldots, N$$

## 2.2 Properties of Equilibrium

When $r$ is set exogenously, wages in all locations are determined by local productivity levels and the global interest rate. This ensures a unique equilibrium, as the distribution of agents in the second period is solely a function of exogenous factor prices and migration costs. Although it is substantially more parsimonious, the two-period model mirrors several outcomes in the neoclassical and economic geography frameworks.

[6] The probability and measure of agents moving from location $i$ to $j$ are equal to $\pi(i,j)$ as each island is populated by a continuum of agents with measure one.

**Lemma 1.** *It is optimal for agents to populate all locations regardless of damages in the second period.*

*Proof.* As $m(i,i) = 0$ by assumption, $W(i,i) \geq k > -\infty$ for all islands as long as the gross interest rate is positive (i.e., $1 + r > 0$). It follows from the fact that $W(i,j)$ is bounded from above for all $(i,j)$ pairs that $\pi(i,i)$ is strictly positive for all $i$. □

The lemma shows that regardless of how extreme the local effects of climate change will be on a given island, there are always agents who will stay even if losses reduce second period labor income to zero. This feature is not an artifact of functional forms; Lemma 1 would hold for an arbitrary distribution of second-period productivities and migration costs or if production technology were linear. In equilibrium, the nature of the extreme value shocks will cause agents move to islands with lower ex-ante conditional values than their original location despite perfect foresight. Imposing agents forgo consumption goods to migrate helps discipline this aspect of the model:

**Lemma 2a.** *If pecuniary migration costs are zero, all destinations receive migrants from all origins unless $U = \infty$.*

*Proof.* When $m(i,j) =$, all destinations are feasible while maintaining positive consumption in both periods, so $W(i,j)$ in (2) is bounded from below for all possible $(i,j)$ pairs unless $U = \infty$. By (4), all $\pi(i,j)$ are then positive. □

Lemma 2a highlights the equilibrium implications of having migration costs come only in the form of additive or multiplicative effects on utility. When the shocks $\lambda(j)$ are drawn from an unbounded support, there's always a share of agents who will find location $j$ optimal. Regardless of how costly migration is in terms of units of utility, there will always be idiosyncratic benefit that outweighs this cost, no matter the modeled distance. This o utcome can be shut off through utility shocks only by imposing a utility cost of negative infinity for a given $(i,j)$ pair.

**Lemma 2b.** *For a given location $i$, if pecuniary migration costs are sufficiently large no agents move.*

*Proof.* Any $m(i,j) \geq y + (1+r)^{-1} w_2(j)$ ensures no agents move from $i$ to $j$ as doing so would lead to non-positive consumption. If this cost is sufficiently large for all destinations $j \neq i$, no agents move. □

Lemma 2b shows that imposing pecuniary migration costs lets us discipline the dispersion result in Lemma 2a. Any migration cost $m(i,j)$ larger than $\bar{y} = y + (1+r)^{-1}w_2(j)$, where $w_2(j)$ is the highest equilibrium second-period wage, would force negative consumption in the first period and thus precludes emigration from island $i$ entirely. In contrast with the utility cost framework

To illustrate the implications of Lemmas 2a and 2b consider the case of pecuniary migration costs from island $i$ being fixed at $m(i,j) = \tilde{m}$ for all potential destinations $j \neq i$ while utility costs are fixed at $U$. Figure 1 shows how increasing utility costs $U$ or pecuniary costs $\tilde{m}$ affects the share of agents $\pi(i,-i)$ who leave the island in equilibrium in green and blue respectively. For the case of raising $U$ (increasing utility costs), the share of emigrants $\pi(i,-i)$ in green will asymptote to zero but never reach it for finite values. In sharp contrast, when the pecuniary cost of emigration $\tilde{m}$ rises past $\bar{y}$, no agents who begin on island $i$ are able to emigrate regardless of how second-period wages there may be (as shown by the blue function). Similar forces, namely the relative sizes of income and migration costs, cause liquidity constraints to restrain migration in the full model. This feature will the quantitative model in Section 3 replicate the large number of zeros in bilateral migration flows between countries observed in the data (Grogger and Hanson 2011) and low emigration rates from poor areas generally despite large wage differences (Lagakos, Mobarak, and Waugh 2023).[7,8]

[7]This is consistent with the finding that emigration poses a substantial financial cost (as opposed to a loss of utility alone) in many countries (Clemens, Montenegro, and Pritchett 2019). This result can be refined by precluding borrowing, which will in general further reduce emigration from poor areas even when wage gaps are large. Other frameworks that model migration costs as utility losses (cf. Cruz and Rossi-Hansberg 2023) typically do not condition these shocks on income and cannot capture this income-dependent constraint. Modeling savings causes the ability to emigrate to be jointly determined by wealth and migration costs.

[8]Zeros comprise over 25 percent of the estimated bilateral migration flow values between the 154 countries I use in the full model. This figure rises to over 50 percent of all flows under more conservative demographic accounting methods (Abel and Cohen 2022).

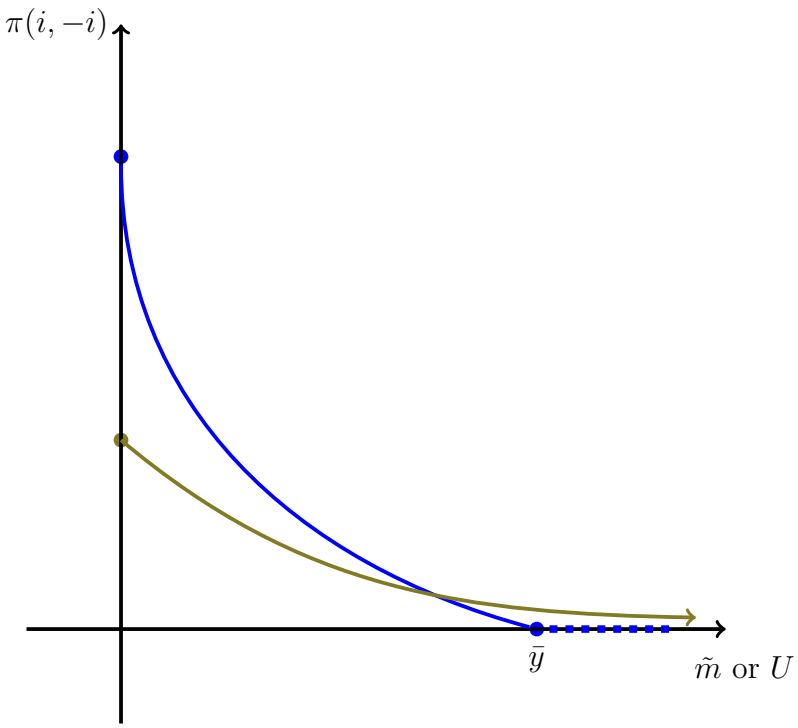



Figure 1: Share of Leavers as a Function of Migration Costs

Incorporating moving costs seems critical for determining the effects of climate change on migration patterns. Evidence from field and natural experiments suggests that high migration costs in tandem with liquidity constraints play a substantial role in explaining why agents remain in low-productivity areas despite existing wage gaps (Bazzi 2017; Cai 2020).[9] Lemma 2b shows how the model allows for these features to shape equilibrium outcomes and can allow different propensities to migrate for poor and rich agents in the same location.

**Lemma 3.** *Imposing a stricter borrowing limit than the natural one (weakly) lowers the share of emigration from all islands.*

*Proof.* Imposing $S(i,j)$ be greater than the NBL term in (1) above can only (weakly) decrease the set feasible destinations from origin $i$. As the numerator in $\pi(i,i)$ is unchanged, it is weakly increasing in stringency. □

Lemma 3 illustrates the important role of limits on borrowing when migration costs enter agents' budgets. When some islands can only be accessed by borrowing against future income

[9]The presence of liquidity constraints fails to explain the dearth of migration in some settings. Lagakos, Mobarak, and Waugh (2023) argue for seasonal migration in Bangladesh that low rates of uptake by poor households are due to it being an expensive form of insurance.

(e.g., when $m(i,j)$ is large compared to $y$ but small compared to $w_2(j)$ wages), tighter limits on borrowing may render these locations inaccessible. Lagakos, Mobarak, and Waugh (2023) show that saving and borrowing to pay migration costs plays a critical role in rationalizing experimental evidence on how migration responds to real income differences. Borrowing limits prove especially important in the quantitative version of the model, where the zero borrowing constraint binds and precludes poor agents in low-income countries from being able to borrow against future (higher) incomes in order to pay up front migration costs.

Comparative statics on how changing migration costs and damages affect the share of agents who migrate are tractable and intuitive:

**Proposition 1.** *Holding $r$ constant, an increase in pecuniary migration costs (weakly) raises the share of stayers $\pi(i,i)$ in all locations. An increase in damages $\theta(i)$ (weakly) lowers the share of arrivals it receives, $\pi(j,i)$, from all islands. The effects are strict outside of the corner case where no emigration occurs.*

*Proof.* See Appendix A. □

When the demand for savings plays no role in determining the equilibrium interest rate, the effect of increasing migration costs is unambiguous; raising costs lowers the net benefit of moving. The share of agents staying on each island between periods one and two increases on all islands and is offset by a decline in emigration to other islands. An intuitive converse holds for the local effects of larger second-period damages. Low idiosyncratic productivity on that island lowers wages while having no effects on wages in all other locations due to the interest rate being fixed. These static results also attain in the limiting economic geography case where savings is prohibited.

**Proposition 2.** *An increase in $r$ raises the share of stayers in all locations with below-average productivity in the second period. The effect is ambiguous for places with above-average productivity, but becomes more likely as migration costs rise.*

*Proof.* See Appendix B. □

Increasing the exogenous return on savings (cost of borrowing) raises the share of agents who stay on the less-fortunate islands. Higher $r$ lowers second period wages across all locations relative to moving costs and makes paying the up-front costs of migration more costly in terms of second period income for agents who borrow to move. The effect is ambiguous for agents on islands with high-incomes in the second period; while all wages fall relative to migration costs, higher $r$ compresses the absolute dispersion in wages across locations and may make the agent less averse to moving despite being on a "good" island. However, when

migration costs are sufficiently high, the first effect dominates on all islands and the increase in relative migration costs causes less emigration everywhere.

## 2.3 Endogenous $r$

Fixing $r$ exogenously lent the simple model clean insights. However, as climate change is a global phenomenon, I consider a refined equilibrium concept from Definition 1 to ensure that the supply of capital meets demand:

4. The capital market clears at rate $r$

$$\sum_{i=1}^{N} K_2^D(i)\, di = \sum_{i=1}^{N}\sum_{j=1}^{N} \pi(i,j) S(i,j) \tag{5}$$

where

$$K_2^D(j) = A_2(j) N_2(j) \left(\frac{\alpha}{r}\right)^{\frac{1}{1-\alpha}} \tag{6}$$

Condition 4 requires that capital supply meets demand and closes the simple model in general equilibrium. Appendix C proves at least one equilibrium exists for this economy. While the Lemmas 1-3 hold in this setting, imposing general equilibrium breaks the clean results presented in Proposition 1. The effect of changing migration costs or $A_2(i)$ in isolation on the share of stayers is ambiguous for general parameterizations.[10] As suggested by Proposition 2, this occurs because any knock-on effects on equilibrium interest rates can have ambiguous effects on the rate of stayers for islands with favorable draws of damages $\theta$. Shifts in equilibrium interest rates caused by increased migration costs may have different effects for agents who borrow in order to move (for whom it costs more to migrate) versus those who gain more from staying in place. This indeterminacy holds even for the case where multiple equilibria are ruled out.[11]

The simple model highlights the importance of space and time as distinct channels for adaptation to changing climate. The model suggests at some point we should expect agents to use real resources to avoid areas worst hit by climate change. However, solely allowing for adaptation through location choice misses the ability for agents to partially insure by substituting intertemporally.[12] Determining outcomes when both channels are available requires modeling the intersection of these two channels for adaptation.

[10] See Appendix D.

[11] Appendix E shows that even in the case where the equilibrium is unique, the effects of changes in migration costs or damages on the share of stayers cannot be signed.

[12] Firms' decisions in Cruz and Rossi-Hansberg (2023), the leading SIAM, do allow for a form of investment

# 3 A Model of Global Savings and Migration

The rest of the paper takes the framework of the two-period model to a calibrated infinite-horizon setting. Time is discrete with each period corresponding to one year. The economy is populated by a large number of agents, each of whom resides in one country, at a given point in time. Each country is modeled as having a representative firm equipped with Cobb-Douglas production technology that is augmented by a country-specific level of labor productivity. Firms must hire local labor, but have access to an integrated capital market that rents out global savings at a uniform risk free rate. Agents' choices each period consist of a traditional consumption savings problem and choosing a (potentially new) location where to work and reside. Outside of preference shocks affecting location choices, agents face idiosyncratic risk over their ability to supply labor, leading to an incomplete-markets model in the tradition of Aiyagari (1994). In a stationary equilibrium, individual agents' locations and asset holdings change over time while the joint distribution of locations, labor endowments, and assets is stationary. This section outlines the model structure and defines an equilibrium concept for the model in a steady-state.

## 3.1 Agents

The world is populated by a continuum of infinitely-lived agents of measure one, each identical in their preferences and discount factors who live in one of $N$ countries each period. Time is discrete. Agents supply labor inelastically to a representative firm in their location which produces a single uniform good that can be consumed or saved in the form of capital. Each agent can supply labor $s$ each period, such that an agent in location $i$ receives labor income $sw_i$. Like in Aiyagari (1994), agents face idiosyncratic risk to their labor supply; labor endowments $s$ follow a Markov process that switches between a high $h$ and low $\ell$ state with transition probabilities $P(s'|s)$ denoted by $\Pi_{ss'}$.

As in the two-period model, each agent receives a set of location-specific idiosyncratic taste shocks each period denoted by $\{\lambda_t(i)\}_{i=1}^N$. Agents must pay $m_t(i,j)$ units of the consumption good in order to move from location $i$ to location $j$ in period $t$. Migration costs are non-negative, potentially asymmetric, and satisfy $m_t(i,i) = 0$. This modeling choice acknowledges that migrants cannot overcome the barriers between borders through pure disutility alone. The mechanism here instead reflects the fact that international migration

in the sense that firms' choose a level of technological innovation each period, but their problem reduces to a static one and this investment does not allow agents to smooth consumption through savings by selling claims on the firms' profits. Other spatial formulations allow for capital accumulation, but the agents that do so cannot migrate (cf. Bilal and Rossi-Hansberg 2023). Agents in Hassler, Krusell, and Olovsson (2026) can move within countries (while saving) but cannot move internationally.

net of physical travel costs, whether it is visa fees or so-called "golden passports", is never entirely free of charge (Recchi et al. 2021; Surak 2021).

Each period, in addition to choosing where to move, agents choose their next-period asset holdings $a_{t+1}$ that supply productive capital to firms in the next period on a frictionless global capital market. Firms in each country rent capital at a risk-free rate $r_t$ net of depreciation. I assume agents' borrowing is constrained by a lower bound $\underline{a}_t$ each period, which may be greater than or equal to the natural borrowing limit conditional on where they choose to move. The set of possible feasible asset holdings $\mathcal{A}_t$ is then the interval $[\underline{a}_t, \infty)$.

Each agent's utility maximization problem is to choose a sequence of decision rules governing savings and location choices—$\{a_{t+\tau+1}, j_{t+\tau+1}\}_{\tau=0}^{\infty}$—conditional on her observed states and taste shocks that maximizes her expected lifetime utility:

$$
\begin{aligned}
V_t\Big(a_t, s_t, j_t, \{\lambda_t(i)\}_{i=1}^N\Big) = \max\ & \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^\tau \Big( \ln(c_{t+\tau}) + \lambda_{t+\tau}(j_{t+\tau+1}) \Big) \qquad (7) \\
\text{s.t.} \quad & s_{t+\tau} w_{j_{t+\tau}, t+\tau} + (1 + r_{t+\tau}) a_{t+\tau} = a_{t+\tau+1} + c_{t+\tau} + m_{t+\tau}(j_{t+\tau}, j_{t+\tau+1}) \\
& a_{t+\tau+1} \in \mathcal{A}_{t+\tau+1} \quad , \quad \forall \tau \\
& c_{t+\tau} > 0, \qquad \forall \tau \\
& \text{with } \Big(a_t, s_t, j_t, \{\lambda_t(i)\}_{i=1}^N\Big) \quad \text{given}
\end{aligned}
$$

subject to her budget constraint, non-negative consumption, her current idiosyncratic shocks, and a sequence of period-specific borrowing limits $\underline{a}_{t+\tau+1}$ which dictate maximal borrowing.

## 3.2 Production

Let $\Omega_t(a, s, j)$ denote the joint distribution of assets, labor endowments, and locations at time $t$. The population of location $j$ is given by the mass of agents living there that period

$$
L_{j,t} \ = \ \int_{a \in \mathcal{A}_t} \sum_{s \in \{\ell, h\}} d\Omega_t(a, s, j), \qquad (8)
$$

which due to normalization is also equal to the share of the global population that resides there. Aggregate labor supply in each location $j$ is then

$$
N_{j,t}^S \ = \ \int_{a \in \mathcal{A}_t} \sum_{s \in \{\ell, h\}} s \ d\Omega_t(a, s, j).
$$

Each location is endowed with a fundamental level of labor-augmenting productivity $\bar{A}_j$ reflecting the large variation in per-capita incomes observed in the data. Productivity

grows at a rate of $g$ each year such that productivity in all locations can be expressed as $A_{j,t} = \bar{A}_j(1+g)^t$. Production is done by a representative firm in each location which operates using a Cobb-Douglas production technology that combines capital $K$ and effective labor as inputs to produce a uniform final good:

$$Y_{j,t} = K_{j,t}^{\alpha}(A_{j,t}N_{j,t})^{1-\alpha}.$$

Firms take the global rental cost of capital $r_t + \delta$ and local wages $w_{j,t}$ as given and choose inputs to maximize profits. Constant returns to scale imply zero profits at equilibrium factor prices, so equilibrium wages may then be written solely as a function of the global interest rate $r_t$ and local fundamental productivity levels:

$$w_{j,t} = A_{j,t}(1-\alpha)\left(\frac{\alpha}{r_t+\delta}\right)^{\frac{\alpha}{1-\alpha}}. \tag{9}$$

Equilibrium capital demand in each location satisfies

$$K_{j,t}^{D} = A_{j,t}N_{j,t}^{D}\left(\frac{\alpha}{r_t+\delta}\right)^{\frac{1}{1-\alpha}}.$$

## 3.3 Recursive Formulation

At this point it is convenient to transform the agent's problem in (7) into a stationary form before expressing it recursively. When productivity growth is constant, the path of endogenous variables can be expressed in normalized terms such that they do not grow without bounds. Define $\gamma \triangleq (1+g)$ such that $A_{j,t} = \gamma^t \bar{A}_j$. Let hat values

$$\hat{z}_t \triangleq \frac{z_t}{\gamma^t}$$

denote the value of a given variable normalized by cumulative productivity growth at period $t$ relative to some initial period 0. I also assume that the borrowing limit scales with productivity, $\underline{a}_t = \gamma^t \hat{\underline{a}}$, so that the normalized asset space is time-invariant: $\hat{\mathcal{A}} = [\hat{\underline{a}}, \infty)$. Towards a stationary equilibrium concept, I assume migration costs grow by the same exogenous factor $\gamma$ each period such that costs relative to time $t$ can be expressed as

$$m_{t+\tau}(i,j) = \gamma^{\tau} m_t(i,j). \tag{10}$$

The agent's sequential problem can be rewritten as

$$V_t\Big(a_t, s_t, j_t, \{\lambda_t(k)\}_{k=1}^N\Big) = \max\ \mathbb{E}_t \sum_{\tau=0}^{\infty} \beta^\tau \Big((\tau + t)\ln(\gamma) + \ln(\hat{c}_{t+\tau}) + \lambda_{t+\tau}(j_{t+\tau+1})\Big) \tag{11}$$

$$\begin{aligned} \text{s.t.} \quad & s_{t+\tau}\hat{w}_{j_{t+\tau},t+\tau} + (1 + r_{t+\tau})\hat{a}_{t+\tau} = \gamma\hat{a}_{t+\tau+1} + \hat{c}_{t+\tau} + \hat{m}(j_{t+\tau}, j_{t+\tau+1}) \\ & \hat{a}_{t+\tau+1} \in \hat{\mathcal{A}}_{t+\tau+1} \\ & a_t, s_t, j_t, \{\lambda_t(k)\}_{k=1}^N \quad \text{given} \end{aligned}$$

Rewrite the function on the left hand side of Equation (11) in stationary form

$$V_t\Big(a_t, s_t, j_t, \{\lambda_t(i)\}_{i=1}^N\Big) = \hat{V}_t\Big(\hat{a}_t, s_t, j_t, \{\lambda_t(i)\}_{i=1}^N\Big) + \frac{t\ln(\gamma)}{(1-\beta)} + \frac{\beta\ln(\gamma)}{(1-\beta)^2}$$

where the last two terms are constants that depend only on the absolute time index of the program $t$. This normalized value function solves the functional equation subject to the normalized budget constraint

$$\hat{V}_t\Big(\hat{a}_t, s_t, i_t, \{\lambda_t(j)\}_{j=1}^N\Big) =$$

$$\max_j \left\{ \max_{\hat{a}_{t+1} \in \Gamma(\hat{x}_t, j)} \left\{ \ln(\hat{c}_t) + \lambda_t(j) + \beta\, \mathbb{E}\Big[\hat{V}_{t+1}(\hat{a}_{t+1}, s_{t+1}, j, \{\lambda_{t+1}(k)\}_{k=1}^N) \mid s_t\Big] \right\} \right\} \tag{12}$$

$$\begin{aligned} \text{s.t.} \quad & (1 + r_t)\hat{a}_t + \hat{w}_{i_t,t}s_t = \gamma\hat{a}_{t+1} + \hat{c}_t + \hat{m}(i_t, j) \\ & i_{t+1} = j \\ & \hat{a}_{t+1} \geq \underline{\hat{a}} \end{aligned}$$

where $\underline{\hat{a}}$ denotes the borrowing constraint for the stationary version of the problem. The maximizer $\hat{a}_{t+1}$ is her (optimal) savings in effective units conditional on a choice of next-period location $j$ where $\Gamma(\hat{x}_t, j)$ is the constraint correspondence induced by the borrowing and budget constraints along with positive consumption given her current state and choice of destination in the next period.[13,14] The policy functions that maximize the right-hand side of equation (12) coincide with the sequence of (normalized) decisions that maximize (11).

Let $\hat{x}_t \triangleq \{\hat{a}_t, s_t, i_t\}$ summarize an agents' state variables (her normalized asset holdings,

[13] Explicitly,

$$\Gamma(\hat{x}_t, j) = \Big\{\hat{a}' \in \hat{\mathcal{A}} : \gamma\hat{a}' < (1 + r_t)\hat{a}_t + \hat{w}_{i_t,t}s_t - \hat{m}(i_t, j)\Big\}.$$

An infeasible destination has an empty correspondence and conditional value $-\infty$.

[14] In a slight abuse of notation I've also swapped the notation such that the $(i, j)$ convention is preserved for describing origins and destinations.

labor endowment, and location) and $\mathcal{X}$ denote the normalized state space consisting of potential tuples $\hat{x}_t$. Let $\hat{d}_t \triangleq \{j, \hat{a}_{t+1}\}$ be the tuple of her choice variables over location and asset holdings next period subject to the constraint $\hat{a}_{t+1} \in \Gamma(\hat{x}_t, j)$. Finally, let $u_t(\hat{x}_t, \hat{d}_t)$ be her utility (log consumption) in state $\hat{x}_t$ when she chooses decision $\hat{d}_t$ leading to consumption level $\hat{c}_t$.[15] Write $\mathbb{E}[\hat{V}_{t+1}(\hat{x}_t, \hat{d}_t)]$ as shorthand for the expected continuation value conditional on the current state and decision. Following Rust (1987), when the location taste shocks are independent draws from a mean-zero Type-I extreme value distribution with scale parameter $\nu > 0$, integrating over future taste and labor shocks gives $\mathbb{E}[\hat{V}_{t+1}(\hat{x}_t, \hat{d}_t)]$ can be written as:

$$\mathbb{E}\Big[\hat{V}_{t+1}\Big(\hat{x}_t, \hat{d}_t\Big)\Big] = \sum_{s' \in \ell, h} \Pi_{ss'} \; \nu \; \ln\left(\sum_{k=1}^{N} \exp\left(\frac{1}{\nu} \max_{\hat{a}_{t+2} \in \Gamma(\hat{x}_{t+1}, k)} \left\{u_{t+1}\Big(\hat{x}_{t+1}, \hat{d}_{t+1}\Big) + \; \beta \, \mathbb{E}\Big[\hat{V}_{t+2}\Big(\hat{x}_{t+1}, \hat{d}_{t+1}\Big)\Big]\right\}\right)\right). \tag{13}$$

Here, $\hat{x}_{t+1} = (\hat{a}_{t+1}, s', j)$ and $\hat{d}_{t+1} = (k, \hat{a}_{t+2})$, as the expected continuation value from making decision $\hat{d}_t$ today given state $\hat{x}_t$ depends on the relative probabilities of choosing each possible (new) destination $k$ tomorrow. The right-hand side averages over potential next-period labor endowments using $\Pi_{ss'}$. For each resulting state, the log-sum-exp term integrates the maximized value over next-period taste shocks, evaluated at the optimal savings policy for each possible destination subject to constraints $\Gamma(\hat{x}_{t+1}, k)$.

Define her savings policy function conditional on choice of location, $g_t(\hat{x}_t, k_t) \triangleq \hat{a}_{t+1}$, which solves the inner maximization problem in (12) for each choice of location and corresponds to the optimal decisions in the sequential problem. The functional form of the shocks also leads to a closed-form solution for the probability an agent in state $\hat{x}_t$ will find it optimal to move to location $j$ in period $t+1$. By a law of large numbers argument, this probability is also the share of agents of type $\hat{x}$ who move to location $j$, and is given by[16]

$$\pi_t(\hat{x}, j) = \frac{\exp\left(\frac{u_t\left(\hat{x}_t, \hat{d}_t^j\right) + \beta \mathbb{E}\left[\hat{V}_{t+1}\left(\hat{x}_t, \hat{d}_t^j\right)\right]}{\nu}\right)}{\sum_{k=1}^{N} \exp\left(\frac{u_t\left(\hat{x}_t, \hat{d}_t^k\right) + \beta \mathbb{E}\left[\hat{V}_{t+1}\left(\hat{x}_t, \hat{d}_t^k\right)\right]}{\nu}\right)} \tag{14}$$

[15] The utility function is time indexed due to its implicit reliance on $w_{j,t}$ and $r_t$.

[16] Technical issues arise when applying a conventional law of large numbers argument to economies characterized by a continuum of agents who experience idiosyncratic shocks governed by realizations of random variables. Uhlig (1996) provides a weaker criterion for convergence in probability such that the large numbers argument invoked here, that migration probabilities induce an equal share of agents to move, holds formally.

where the notation $\hat{d}_t^j \triangleq \left\{j, g_t(x_t, j)\right\}$ denotes the vector pairing her location decision with the associated optimal savings level. These probabilities along with a joint distribution of wealth and labor endowments across locations determines the evolution of the distribution of agents. Given a distribution of agents $\hat{\Omega}_t(\hat{x}) \triangleq \hat{\Omega}_t(\hat{a}, s, i)$ across locations, the distribution of agents of type $\hat{x}' = (\hat{a}', s', j')$ at time $t+1$ is given by

$$\hat{\Omega}_{t+1}(\hat{x}) = H_t\Big(\hat{\Omega}_t(\hat{x}_t)\Big) \tag{15}$$

where $H_t(\Omega)$ is the law of motion for the distribution of agents induced by the migration probabilities $\pi_t(\hat{x}, j)$ and policy functions $g_t(\hat{x}, j)$ that period (Ljungqvist and Sargent 2018). I now turn to defining a stationary equilibrium for this economy, where $\hat{\Omega}(\hat{x})$ is a fixed point of (15) and the normalized value functions and conditional choice probabilities in Equations (13) and (14) are time-invariant.

**Definition 2.** A stationary recursive competitive spatial equilibrium consists of a set of expected value functions $\mathbb{E}\Big[\hat{V}(\hat{x}, \hat{d})\Big]$, savings policies $g(\hat{x}, j)$, migration probabilities $\pi(\hat{x}, j)$, an interest rate $r$ and effective wages at each location $\{\hat{w}_i\}_{i=1}^N$, and a joint distribution of normalized assets and labor endowments across locations $\hat{\Omega}(\hat{x})$ such that:

1. Given prices $(r, \{\hat{w}_i\}_{i=1}^N)$, savings policies $g(\hat{x}, j)$ and $\mathbb{E}\Big[\hat{V}(\hat{x}, \hat{d})\Big]$ solve functional equation (13) and induce the probabilities in (14)

2. Given prices $(r, \{\hat{w}_i\}_{i=1}^N)$, firms' factor demand $\hat{K}_j^D$ and $N_j^D$ are optimal in all locations

3. The global capital market clears at interest rate $r$

$$\sum_{j=1}^{N} \hat{K}_j^D \;=\; \int_{\hat{x}\in\mathcal{X}} \sum_{j=1}^{N} g(\hat{x}, j)\; \pi(\hat{x}, j)\; d\hat{\Omega}(\hat{x})$$

4. Labor markets clear at effective wage $\hat{w}_j$ in each location $j$

$$N_j^D \;=\; \int_{\hat{a}\in\hat{\mathcal{A}}} \sum_{s\in\{\ell,h\}} s\; d\hat{\Omega}(\hat{a}, s, j), \qquad j = 1, \ldots, N$$

5. The joint distribution of agents over asset holdings, labor endowments, and locations $\hat{\Omega}(\hat{x})$ is stationary in that it is a fixed point of the transition function $H(\Omega)$ induced by the stationary migration probabilities $\pi(\hat{x}, j)$ and policy functions $g(\hat{x}, j)$

In a stationary equilibrium, the economy behaves as a model of discrete choice over locations where each location choice nests a standard consumption/savings problem with uninsurable risk. When the economy is parameterized with only one region it nests a version of Aiyagari (1994) with two states of possible labor supply. Appendix **??** briefly discusses the numerical issues with this type of dynamic model where agents face discrete and continuous choices (Iskhakov et al. 2017) as well as how multiple equilibria issues in Aiyagari economies (Walsh and Young 2024) may be exacerbated in my setting where aggregate effective labor supply is also determined by the equilibrium distribution of agents.

## 3.4 Discussion

I adopt an incomplete markets model (as opposed to one where wealth dispersion is generated solely by migration decisions) because differentiated wealth and idiosyncratic income shocks have been shown to be key determinants of individuals' decisions over international migration (Gröger and Zylberberg 2016; Bazzi 2017). As the process governing shocks is independent of location and wealth levels, it is intended to capture a broad notion of uninsurable income risks agents face around the world that may interact with decisions to migrate. In low-income regions where a large share of workers are employed in agriculture, these shocks may be interpreted as household-specific income exposure to weather fluctuations, which have been shown to drive migration and labor reallocation (Colmer 2021; Barbosa-Alves and Braulio 2023). However, these shocks may also comprise those affecting employment more broadly, as declines in local labor demand can lead to internal migration in settings beyond the agricultural sector (Cai 2020).

More generally, when the costs of migration enter agents' budget constraints, migration rates in a given region can vary across the local income distribution. As illustrated by Lemma 2b in the simple model, this is because the set of agents who face binding constraints on migration is determined by wealth and local wages. This reflects that international emigration is increasing in income at low income levels within countries and in the cross-sectional data (Cai et al. 2016; Bazzi 2017) because of factors such as credit constraints and large up-front costs (Clemens, Montenegro, and Pritchett 2019; Lagakos, Mobarak, and Waugh 2023). Both of these frictions can become barriers to migration in the model; the set of agents in a given origin region for whom migration costs are prohibitively high is determined endogenously in equilibrium.

# 4 Calibration, Solution, and Estimation of the Baseline Economy

This section takes a version of the model with $N = 154$ regions, each corresponding to a country, to the data. Solving the model and calibrating a baseline requires inputs for the structural parameters governing the model, namely the fundamental productivity and initial temperature levels in each country, bilateral migration costs between each pair of countries, and the deep parameters governing production and preferences. After a subset of parameters is externally calibrated, the remaining ones are chosen to minimize the distance between the shares of the modeled global population in each country and the observed shares in the data. Sections 4.1 and 4.2 list the parameters that I set directly based on external sources: those taken from the literature or set directly based on observable analogs of model objects. Section 4.3 describes the solution algorithm for the model given a set of structural parameters. Section 4.4 lists the remaining free parameters in the model that I estimate using the method of simulated moments described in Section 4.5.

## 4.1 Calibration: Parameters Sourced from Existing Estimates

**Fundamental Productivity and Population Levels:** Fundamental baseline productivity levels are assigned based on the 2010–2014 average of real per capita GDP as recorded in the World Bank's World Development Indicators (WDI – WBG 2016) database. I normalize this measure of labor-augmenting productivity such that all values are relative to that of the United States in 2010. The target moments for the baseline economy are based on each country's average share of the global population over the same five-year period in the WDI database.

**Projections of Future Climate Change:** I take projections for temperature trajectories between 2015 and 2100 using output from the Coupled Model Intercomparison Project Phase 6 (CMIP6). The Coupled Model Intercomparison Project provides a consensus on a set of standardized experiments that allow climate modelers to coordinate the design and comparison of output on a harmonized basis (Eyring et al. 2016). I focus on two benchmark scenarios for anthropogenic greenhouse gas emissions, Representative Concentration Pathways (RCPs) 4.5 and 8.5, which are indexed numerically according to the associated increases in radiative forcing by 2100 (Van Vuuren et al. 2011).[17,18] The RCPs are a set

[17] I am grateful to Jamie Scott at the University of Colorado for facilitating an easy file transfer process for accessing the data containing these ensemble averages.

[18] Using only the climate projections from CMIP6 leads to a minor internal inconsistency, as the two RCPs I consider are paired with different Shared Socioeconomic Pathways, which embed assumptions about the

of paired socioeconomic and greenhouse gas emissions trajectories between 2005 and 2100 (Van Vuuren et al. 2011), each guided by a qualitative narrative for how the world develops as a whole over the coming century (Lee et al. 2021). Higher indices indicate larger increases in forcing and, with it, cumulative warming. To form RCP-specific projections of future temperatures through 2100, I use the simple average of projected annual warming across 33 models participating in CMIP6. The projections provide annual averages of near-surface temperatures at a $1^\circ \times 1^\circ$ resolution globally for each of the two RCPs I consider. I then aggregate the grid-scale temperature data into population-weighted average annual temperatures in each country using gridded population data (base year 2015) from the CIESIN Gridded World Population v4 dataset (GPWv4 – CIESIN 2017) and boundaries from the Global Administrative Database (GADM 2022).

**Historical Temperatures:** I use a process analogous to that for future temperatures to construct population-weighted average temperatures at the country level. Baseline temperatures in each country, $\bar{T}_j$, are constructed as arithmetic averages of population-weighted temperatures between 1955 and 2014, where time-invariant weights are again assigned using the GPWv4 data (CIESIN 2017).

**Climate Damages:** Nath, Ramey, and Klenow (2023) use local projection methods along with country-level panel data on temperature and GDP to estimate country-specific response functions for how climate change affects output growth.[19] These projections combine to form estimates of the cumulative effects of temperature increases on output growth for each country in the panel they use for estimation. I use these cumulative response functions to fit a parameterized function $D(\bar{T}_j, \Delta T_{j,t})$ that maps country-level historical average temperatures, $\bar{T}_j$, and future changes in local temperatures due to climate change, $\Delta T_{j,t}$, into contemporaneous effects on productivity growth each year. For country $j$ that experiences an increase in average annual temperature of $\Delta T_{j,t}$ in year $t$, the resulting effects on productivity growth are given by

$$\frac{\Delta A_{j,t}}{A_{j,t}} = g + \frac{D(\bar{T}_j, \Delta T_{j,t})}{1-\alpha} \tag{16}$$

where the $1-\alpha$ term in the denominator of the second term ensures that the damages to labor-augmenting productivity in my model translate to the estimated losses in output growth found in NRK, holding effective labor and capital constant.

paths of economic growth, $g$. More precisely, my experiments examine the effects of climate change projected under the emissions levels for a given SSP-RCP pair.

[19] I thank Ishan Nath for providing the point estimates from Nath, Ramey, and Klenow (2023).

## 4.2 Calibration: Directly Chosen Parameters

**Labor Supply Shocks:** I normalize the high labor productivity level $s_h$ to one and set the low productivity state $s_\ell$ to 0.2. This level is halfway between zero (a complete loss of labor income) and the level of 0.4, which is a common parameterized value for unemployment benefits (Shimer 2005). I set $\Pi_{\ell h}$ and $\Pi_{h\ell}$, the transition probabilities into and out of the high state, respectively, to match an equilibrium share of about 5 percent of agents in the low state and an expected bad shock duration of two years.

**Borrowing Constraints:** I set the lower bound on normalized asset holdings to 0, precluding borrowing. Three factors drive this decision. First, because per capita wages are exceptionally low in many countries relative to global averages, the natural borrowing limit is already very close to zero for agents who live in low-income countries or choose to migrate there. Second, imposing state- and location-specific natural borrowing limits requires tracking state-contingent borrowing limits for each agent during simulations, which is computationally expensive. Finally, in models of low-income settings, it is common to assume that commitment is asymmetric and to prevent agents from borrowing even when they have access to risk-free savings (e.g., Lagakos, Mobarak, and Waugh 2023).

**Deep Parameters:** All other parameters that are chosen externally are shown in Table 1, which displays their values as well as their sources in the literature. These parameters govern pure time preference, the capital share in production, the depreciation rate of capital, long-run productivity growth, and the sensitivity of migration to changes in values of different locations.

Table 1: Externally Calibrated Parameters

| Parameter | Value | Source |
|---|---|---|
| $\alpha$ | 0.3 | Hassler, Krusell, and Olovsson (2026) |
| $\beta$ | 0.985 | Hassler, Krusell, and Olovsson (2026) |
| $\delta$ | 0.06 | Krusell and Smith (2022) |
| $g$ | 0.01 | Piketty (2014) |
| $s_h$ | 1 | Normalized |
| $s_\ell$ | 0.2 | Based on Shimer (2005) |
| $\underline{a}$ | 0 | Lagakos, Mobarak, and Waugh (2023) |
| $\nu$ | 1 | Standard |

## 4.3 Solving the Model

Solving the model in general equilibrium requires stationary value and policy functions along with an interest rate at which the global capital market clears. The method for a given set

of structural parameters is fairly standard for incomplete markets models (e.g., Ljungqvist and Sargent 2018) outside of the approach to the value and policy functions. The rest of this Section focuses on methodology and a reader more interested in the goodness of fit for the baseline model and/or results from policy experiments can skip directly to Section 5.

I use simulation-based methods to solve for the ergodic distributions induced by a given optimal value and policy function given the high dimensionality of the transition probabilities and potential limited sparsity around gridpoints where migration costs are lower than wealth levels. For a given set of calibrated parameters, deep parameters (Table 1), and remaining parameters estimated in Section 4.5, the solution algorithm is as follows:

#### 4.3.1 Aiyagari Algorithm

1. Set outer loop iterations to $i = 0$ and initialize a guess of the initial equilibrium interest rate $r^i$ in the open interval $r^i \in (-1, \frac{\gamma-\beta}{\beta})$.

2. Use value function iteration (VFI) to solve for value functions $\mathbb{E}[V(\cdot)|r^i]$ that satisfy (13) conditional on $r^i$.

3. Solve for the resulting conditional choice probabilities $\pi(\hat{x}, j|r^i)$ induced by the conditional value functions $\mathbb{E}[V(\cdot)|r^i]$.

4. Initialize 100,000 agents with random values for initial state $\hat{x}$.

5. Draw a realization from $\mathcal{U}[0, 1]$ for each agent and map it into a (feasible) choice of next-period location and solve their associated optimal consumption/savings decision.

6. Step the Markov process governing labor endowments forward for each agent and combine it with decisions $\hat{d}$ to form the new state tuples $\hat{x}'$.

7. Repeat steps 5-6 until the cross-sectional distribution of locations and wealth has settled (roughly 50,000 time periods).

8. Solve for the interest rate implied by aggregate savings along with aggregate (effective) labor supply:

$$\tilde{r}^i = \alpha K^s(r)^{\alpha-1} \left( \sum_{j=1}^{N} A_j N_j \right)^{1-\alpha} - \delta.$$

9. If absolute value $|\tilde{r}^s - r^i|$ is below a tolerance threshold ($10^{-4}$), set $r^* = \frac{\tilde{r}^s + r^i}{2}$ and stop.

10. If not, set $r^{i+1} = \zeta\tilde{r}^i + (1-\zeta)r^i$ where $\zeta \in (0,1)$ is a dampening parameter and repeat steps 2-8.

There are a few aspects of the problem here that makes operationalizing this algorithm difficult. The first comes with having to iterate Step 2 many times given the size of the state space and nature of the choice set. Solving for optimal conditional consumption and value functions in a model with discrete and continuous choice variables is substantially more difficult than in pure discrete choice problems. A continous choice variable means optimizing over asset holdings at each state space node. The discrete aspect means the value functions in equation (12) may not be concave, which affects optimal savings rules even in the presence of extreme value shocks that allow for migration choices to be probabilistic (Iskhakov et al. 2017). In my case, this is qualitatively important, as value functions differ greatly across locations and budget constraints frequently determine the set of feasible destinations. Accurately approximating the nonlinearities in policy functions governing migration due to borrowing constraints is critical for the numerical results to correctly account for who stays and leaves.

The computational difficulty in this problem is ensuring that my numerical solution captures the nonlinearities in savings policies by accommodating a fine asset grid while preserving computational tractability. Because of the large dispersion in productivity levels across countries, the asset grid should contain many nodes near $a = 0$ to ensure I capture the concavity in the savings behavior of low-income agents. However, each node in a traditional asset grid for value function iteration increases the number of conditional value functions by about $10^5$. This requirement for a large number of nodes quickly renders the benchmark VFI approach, where interpolated value functions and numerical differentiation are used for selecting optimal savings policies, computationally infeasible. Iskhakov et al. (2017) shows that an extension of the endogenous grid method first proposed in Carroll (2006) can substantially improve performance in this dynamic discrete choice setting while preserving a continuous choice set. Appendix F lays out this approach and explains its advantages over discretization while remaining faster than VFI with continuous choices; I have not yet been able to implement the method from Iskhakov et al. (2017) at a speed that renders estimation feasible. I opt instead to use a discretized asset grid that allows the inner optimization problem in (13) to be solved using a single broadcast maximization operator. As pointed out by Iskhakov et al. (2017), this can produce issues when value functions have kink points or the grid used is sparse. This is especially important in my setting if the grid is too coarse in regions where agents seek to build sufficient wealth to migrate. Work implementing the alternative approach described in Appendix F at scale is ongoing in order to ensure that the discretized method is not resulting in large approximation errors.

The second issue is the potential for multiplicity of equilibria even with numerically accurate approximations of value and policy functions in hand. Walsh and Young (2024) show the existence of multiple equilibria for a large set of parameterizations of Aiyagari economies. I cannot apply the results for large economies from Acemoglu and Jensen (2015), as the extreme value shocks are not drawn from a bounded support. While the risk aversion coefficient with log utility and the depreciation value $\delta$ I select lie within the parameter space where Walsh and Young (2024) do not find multiple equilibria, it is possible that multiple equilibria can exist holding a given interest rate fixed and that accordingly the transition path results depend on the initial equilibrium.

This form of multiplicity holding interest rates constant may occur, for instance, if there is not a unique invariant measure of agents across locations given a fixed interest rate $r$ that induces an aggregate savings level and effective labor supply such that capital supply is equal to demand. This would be the case if the transition function induced by $\mathbb{E}\left[\hat{V}(\hat{x}, \hat{d})|r\right]$ does not satisfy the monotone mixing condition in Hopenhayn and Prescott (1992). Evidence for this form of multiplicity would be hysteresis in the simulated steady state of the model; the distribution that is produced after a large number of simulated periods would depend on how the model is initialized in terms of agents' locations and asset holdings. I do not find evidence for this when iterating steps 5-6 of the Aiyagari algorithm above. However, I cannot guarantee that this is true generally, as opposed to being an object of approximation error.

## 4.4 Estimation: Parameters to Estimate

Section 4.3 describes solving the model conditional on a set of structural parameters. I estimate the remaining parameters, those not assigned in Sections 4.1 and 4.2, governing the model using the method of simulated moments. These include the set of pairwise migration costs between all countries – the set of $m(i, j)$ terms – along with three additional structural error parameters ($\delta_C, \delta_I$, and $\delta_H$), which substantially aid model fit.

**Structural Error Terms:** The structural error terms allow the modeled results to be reconciled with two facts in the data: (1) a vast majority of agents do not migrate in a given year; and (2) despite relatively low wages, a large portion of the world's population lives in China and India. To account for these facts, I add three additive utility shifters to the model (the structural error terms), which give additional utility each period to agents living in China, agents living in India, and agents who choose not to migrate, denoted $\delta_C, \delta_I$, and $\delta_H$, respectively. Agents who reside in China (India) receive an additional $\delta_C$ ($\delta_I$) units of utility that period, and agents who do not to move receive additional utility of $\delta_H$.

**Migration Cost Matrix:** The set of pairwise bilateral migration costs is the only remaining model input unavailable in the data. While estimates of the implicit monetary barriers to international migration exist for some origin-destination pairs (e.g., Clemens, Montenegro, and Pritchett 2019), there does not yet exist a joint estimate of migration costs pairs for the 154 countries I consider jointly. Instead, I allow the entries in this matrix to be free parameters to be estimated such that the stationary equilibrium of the model absent climate change matches the shares of the global population living in each modeled country that we see today.

Estimating each of the $N(N-1) = 23,562$ bilateral migration costs as separate free parameters would be computationally intractable and would require calibrating the model directly on observed bilateral migration moments from the data. To reduce the number of free parameters, I adopt a functional form for migration costs that is based on the income quartile of the two countries along with the Euclidean distance between their population-weighted coordinate centroids. Migration costs to go from country $i$ to country $j$ consist of four components:

$$m(i,j) = m^o(q_i) + m^d(q_j) + \bar{m}_1\rho(i,j) + \bar{m}_2\rho(i,j)^2 \ , \ \forall i \neq j \tag{17}$$

where, in a slight abuse of notation, the values $m(i,j)$ in equation (17) from here forward denote the time-invariant values on the right hand side of equation (10).

The origin and destination fixed effects, $m^o(q_i)$ and $m^d(q_j)$, are determined by the income quartiles $q_i$ and $q_j$ of the origin and destination countries. Each possible origin country is assigned a fixed emigration cost across all possible destinations, $m^o(q_i)$, based on the income quartile of that country in the 2010-2014 period. The same procedure assigns a cost to immigrate to each country, $m^d(q_j)$, based on its income quartile. The third and fourth terms are an additive cost that is a function of distance between the origin and destination countries, parameterized by a quadratic function in the Euclidean norm between the countries' population-weighted centroids $\rho(i,j)$. This allows migration costs to reflect the fact that, holding income levels constant, bilateral population flows decrease in distance traveled in the data (Abel and Cohen 2022). This substantially reduces the number of free parameters while allowing for heterogeneity in bilateral migration costs at the country level and for asymmetric bilateral moving costs between all country pairs.

## 4.5 Estimation: Method of Simulated Moments

I estimate the free parameters in the model that govern migration costs, and the structural error terms using the Method of Simulated Moments (MSM). This approach searches for the

set of 13 parameter estimates

$$\Theta = \left\{ \{m^o(q)\}_{q=1}^4, \{m^d(q)\}_{q=1}^4, \{\bar{m}_i\}_{i=1}^2, \delta_C, \delta_I, \delta_H \right\} \tag{18}$$

that minimizes the distance between the share of modeled agents living in each country predicted in equilibrium and the observed shares of the global population residing in each country between 2010 and 2014.[20] The estimates are parameter values that minimize the gap between observed shares of the global population in each country – $\tilde{L}_j$ – and the equilibrium population shares in the model – $\bar{L}_j(\theta)$ from equation (8) – when it is parameterized by a given candidate vector $\theta$. Modeled shares $\bar{L}_j(\theta)$ are calculated for each candidate $\theta$ using the stationary distribution it induces, $\Omega^\theta(a, s, j)$, and by evaluating the share of the population residing in each country in equilibrium:[21]

$$\bar{L}_j(\theta) \;=\; \sum_a \sum_s \Omega^\theta(a, s, j) \tag{19}$$

The targeted moments are the distances between the modeled population shares in equilibrium, $\bar{L}_j(\theta)$, in each country and the shares of the global population in each country, $\tilde{L}_j$, that we observe in the data:

$$z_j(\tilde{L}_j, \theta) \;=\; (\tilde{L}_j - \bar{L}_j(\theta)). \tag{20}$$

When the model is correctly specified, the MSM identifies the parameters governing the population shares we see in the world, $\tilde{L}_j$, along with the aggregate rate of migration. This relies on the assumption that the moment functions in (20) satisfy

$$\mathbb{E}\Big[ z_j(\tilde{L}_j, \Theta) \Big] \;=\; 0 \tag{21}$$

for some unique vector of parameters $\Theta$ that governs the true data-generating process. The objective contains 154 moments, of which 153 are linearly independent because the population shares sum to one. The MSM attempts to minimize the generalized norm between model and data moments:

[20] An identification issue emerges here because the origin and destination fixed effects are only jointly identified up to an additive constant. The estimation routine allows all eight parameters to be free and restricts them to be non-negative.

[21] After iterating the full solution algorithm on a coarse grid initially, I hold the interest rate fixed, rather than solving for an equilibrium interest rate that clears the aggregate capital market, when running the optimization routine that solves (22). For each candidate parameter vector, I solve the household problem and compute its induced stationary distribution conditional on this interest rate and use this to calculate each $z_j(\tilde{L}_j, \theta)$ in (20).

$$\hat{\Theta} \;=\; \underset{\theta}{\operatorname{argmin}}\; \mathbf{z}(\tilde{L},\theta)'\, W\, \mathbf{z}(\tilde{L},\theta) \tag{22}$$

where $\mathbf{z}(\tilde{L},\theta)$ is a vector of each moment condition from equation (20), and $W$ is the identity matrix. In a slight abuse of notation, the estimate $\hat{\Theta}$ is the set of migration cost parameters and structural effects that calibrate the model to the period ranging from 2010 to 2014. After an initial coarse grid search, a Matlab limited-memory BFGS solver running gradients in parallel takes about 48 hours to converge when run on a 64-core large memory HPC node.[22]

# 5 Estimation Results

Figure 2 gives a visual display of the calibration errors when evaluated at $\hat{\Theta}$. These values measure how well the model replicates the observed shares of the global population residing in each country during the 2010–2014 period. For each country, the map displays the value of $z_j(\tilde{L}_j,\hat{\Theta})$, measured as the percentage-point (p.p.) difference between modeled and observed population shares. Countries with a darker red (blue) shade are those where the model underpredicts (overpredicts) the steady-state share of the global population residing in that country relative to the data. The $R^2$ is 0.78; the model accounts for about three-quarters of the observed variation in the dispersion of the world's population across countries.

The model has trouble rationalizing the population shares of low- (high-) productivity countries that have large (small) populations. The largest errors (in magnitude) are in China and India, where, despite the addition of the structural error terms $\delta_C$ and $\delta_I$, the model undershoots steady-state population shares by three and four percentage points, respectively.[23] Similar issues emerge for Indonesia and Pakistan (the fourth and fifth largest countries in the world by population), where the model struggles to match the large populations relative to productivity levels. Conversely, the model overpredicts steady-state populations in most of Northern Europe;modeled population shares in the Netherlands, Austria, Switzerland, Denmark, Norway, and Sweden all exceed their data analogs by between two and three percentage points due to high productivity levels, small baseline populations, and low distances from neighboring high-productivity countries. This latter feature, namely the overprediction of initial population shares in Europe, proves quantitatively important when evaluating aggregate welfare changes from climate change in the sections below because it assigns too much weight to modeled agents in Europe who gain from warming.

[22]I thank Fuzzy Rodgers and Paul Weakliem UC Santa Barbara and Mitchell Horn at Yale University for helping with my use of the Pod and Bouchet HPC clusters for this project.

[23]For reference, China and India each have about 20 percent of the world's residents.

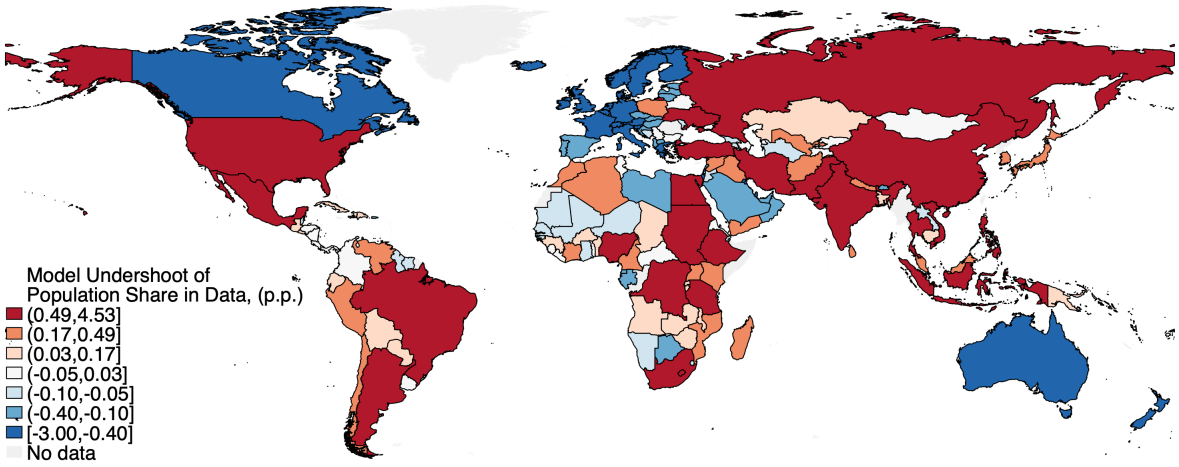



Figure 2: Baseline Model Calibration Errors, (p.p.)

Table 2 displays several migration cost pairs generated by the estimated parameter values $\hat{\Theta}$ that govern the steady-state population levels and migration shares in the calibrated model. To give a sense of scale, I report migration costs as the ratio of the estimated $\hat{m}(i,j)$ value to $\hat{w}_{USA}$ – the equilibrium (stead-state) normalized wage in the United States. I also include the costs and the corresponding home wages in the origin countries reported in dollar terms when the modeled values are converted into \$2019 U.S. dollars using the WDI data (WBG 2016). The costs of migration are lowest between geographically close countries with similar income levels; fitted values imply that it costs about 16% of annual U.S. wages to migrate between Benin and Togo (neighboring small countries in West Africa) or between Norway and Sweden (neighboring medium-sized countries in Northern Europe). This translates to about \$6,900 in 2019 U.S. dollars. As shown in the last column, modeled emigration costs from low-income countries are often substantially higher than (average) wages there.

The estimated costs of moving between countries separated by larger distances or income gaps are even higher. The modeled cost of migration from Guatemala to the United States is 19% of U.S. wages (roughly \$8,000, four times wages in Guatemala), while the extreme case of migrating from Bangladesh to Switzerland has an estimated cost of 28% of U.S. wages (over 20 times wages in Bangladesh). While it is not a direct comparison in terms of the objects identified by the model, the magnitudes of the costs of emigrating from poor to rich countries are similar to implied barriers to migration found in the empirical literature. Clemens, Montenegro, and Pritchett (2019) estimate barriers to low-skilled migration for a panel of 42 low-income countries relative to the United States and find implied costs of between \$5,000 and \$24,000 annually. Bazzi (2017) reports that migration costs from Indonesia to Southeast

Asia and the Middle East through temporary work programs were $1,250 in 2019 dollars.[24] In summary, the calibrated model aligns with the empirical literature suggesting financial barriers are prove prohibitively high.

Table 2: Example migration costs as share of modeled U.S. wages

| Origin-Destination Pair | Migration Cost as Share of Model U.S. Wage | Implied Cost (2019 USD) | Origin Wage $\hat{w}_i$ (2019 USD) |
|---|---|---|---|
| Benin → Togo | 0.160 | $6,900 | $530 |
| Norway → Sweden | 0.164 | $7,026 | $35,630 |
| Guatemala → USA | 0.187 | $8,064 | $2,184 |
| Bangladesh → Switzerland | 0.284 | $12,247 | $568 |

Figure 3 shows how the results in Lemmas 2a and 2b (illustrated with Figure 1 in the stylized model) manifest in the calibrated version. Each line shows the share of agents who emigrate from a given country in the calibrated model, as generated by the migration policy functions:

$$\Pr(\text{Stay in country } i \mid \hat{x}) = 1 - \sum_{j \neq i}^{N} \pi(\hat{x}, j)$$

across varying levels of normalized wealth $\hat{a}$ for four different origin countries holding labor supply fixed. As in the stylized case, as an agent's wealth (not including wages) approaches zero, the probability that the agent stays in place approaches one; agents with no wealth cannot emigrate as they cannot afford the pecuniary costs. This proves critical for determining emigration in countries where income is low; the blue and green lines in Figure 3 show these probabilities, respectively, for two low-income countries in very different geographies, Togo and Haiti. Over half of the agents who reside in each of these countries in the steady state of the calibrated model cannot migrate because they are unable to borrow enough to pay the up-front costs. Their only avenue for adaptation is then either to increase savings locally so as to smooth income in place or eventually to accumulate enough wealth to meet the costs of moving.

[24] I inflate these levels from the values in Bazzi (2017) using the PCE implicit price deflator for the United States so as to provide a rough comparison between values in real terms.

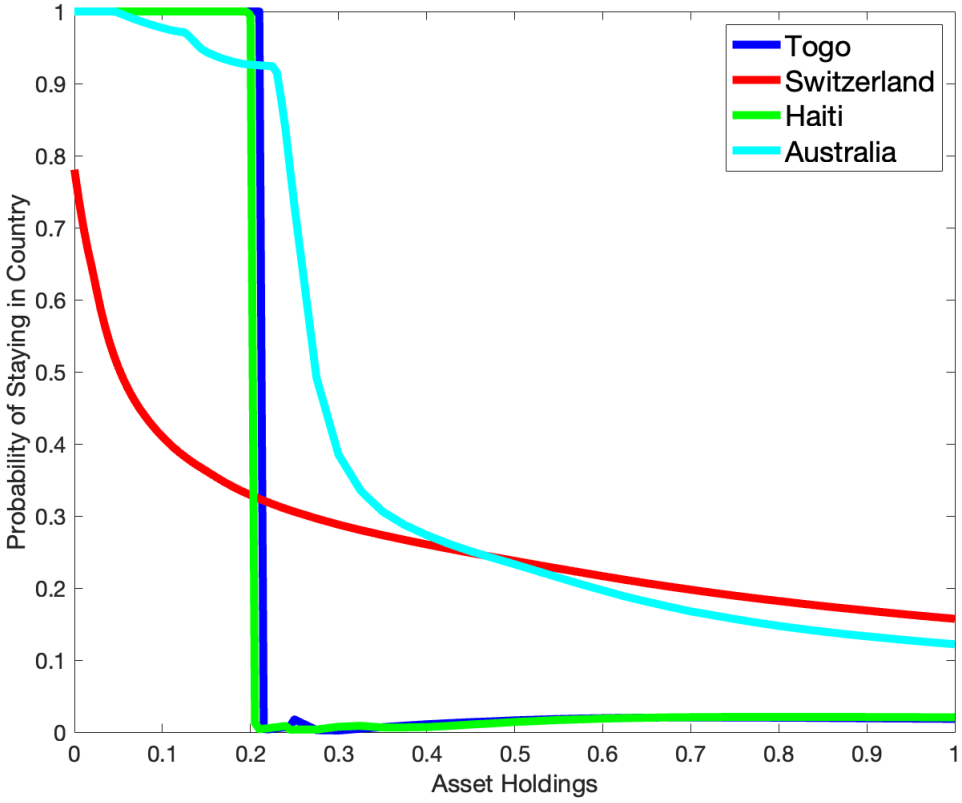



Figure 3: Migration Probabilities as a Function of Wealth in Poor and Rich Countries with Low and High Migration Costs

A converse result is generated for rich agents regardless of moving costs. The model substantially overpredicts the propensity of wealthy and high-income agents to move, as shown by the probabilities of staying in place for Australia and Switzerland (the cyan and red lines in Figure 3). While higher migration costs for Australians, due to the large distance between Australia's population-weighted centroid and those of neighboring countries, do lead to some constraints on migration that are not felt by modeled Swiss agents who enjoy proximity with many other high-income countries, the probability of staying in place falls to about 10 percent for agents in both countries when evaluated at the median wealth leve. However, while the model almost certainly displays too much movement among wealthy agents in response to their taste shocks, this is less likely as agents become poorer, and impossible for constrained agents. Thus when the model is used to forecast population dynamics as the world warms, for the agents for whom borrowing constraints and up-front costs are substantial barriers to using international migration as a form of adaptation, I am less likely to overpredict their response through migration.

# 6 Projections under Climate Change

The calibrated model serves as a laboratory to examine how the effects of climate change on productivity will shape global population dynamics during the 21st century. This section uses steady-state comparisons and transition path exercises to examine these effects at various time horizons. I first use the steady-state comparison framework to perform three experiments exhibiting the relative importance of damages and migration costs in determining the *long-run* effects on the global population distribution. Each experiment solves for a new steady state of the global economy after the world has adjusted to the warming induced through 2100 by a given emissions trajectory. The first, which I refer to as the baseline forecast, examines the change in the steady state of the global economy caused by the roughly 2.0 °C of warming under RCP 4.5. The second re-examines what the steady state would look like in 2100 under the warming induced by RCP 8.5 (approximately 4.2 °C on a population-weighted basis). Finally, I examine the extent to which the new steady state changes if, in addition to the productivity effects of warming under RCP 4.5, migration costs are doubled globally.

The transition path portion of the analysis focuses instead on the short- and medium-run effects of climate change in the modeled economy. This generates a forecast for global population dynamics for the entire 21st century as the world warms and adjusts towards a new steady state. In the transition analysis, each agent in the economy learns in the first year of the transition path that future climate change under RCP 4.5 will alter the trajectory of global productivity levels over the next 85 years and induce a new long-run level thereafter. Each period, agents make new migration and savings decisions according to their new expectations for how climate change will affect future wages and interest rates as these changes become more pronounced with intensifying warming. Because of the slow onset of warming, the experiment differs somewhat from traditional transition paths between stationary equilibria in the macroeconomics literature (cf. Conesa and Krueger 1999) as agents act according to the expected future changes induced by climate change before much of the damage is realized. This exercise (unlike the steady-state comparisons) allows for explicit calculation of how long the world takes to adjust to the new steady state as well as the individual-level welfare effects of climate change across the full distribution of modeled agents.

## 6.1 The Long-Run Effects of Warming

The first counterfactual experiment focuses on how the short- and long-run global distribution of where people live will be affected by climate change in the 21st century. I assume

the new long-run climate under each RCP is determined by the cumulative level of warming predicted by the ensemble average of projections from the Coupled Model Intercomparison Project Phase 6 (CMIP6) for 2100 (Eyring et al. 2016). I focus on RCP 4.5, a "middle-of-the-road" scenario for emissions that induces a relatively modest increase in forcing relative to that of RCP 8.5, which is generally recognized as a pessimistic path for future global emissions. I use it as a baseline here not as a "business-as-usual" scenario for emissions—RCP 8.5 currently tracks most closely with emissions to date among the five central scenarios used by the IPCC (Schwalm, Glendon, and Duffy 2020)—but rather as one that reflects some degree of mitigation implemented at the global scale over the next century. However, analogous results for the effects of RCP 8.5 as opposed to the baseline scenario for local warming and productivity growth are given in parenthetical values throughout this section where applicable.

**Warming:** Figure 4 displays a map of projections from the CMIP6 ensemble average for mean anthropogenic climate change (ACC) projected to occur by 2100 across the globe under RCP 4.5 (RCP 8.5).[25] Each country is colored according to the difference in average annual temperatures between the calibrated 2010-2014 period and the mean forecast across the CMIP6 model outputs for 2100. Warming increases with latitude, with countries in Europe and North America projected to warm the most while southern portions of Africa and South America experience less pronounced temperature changes. Under the simulations based on RCP4.5, the world warms by a projected 2.0 °C (4.2 °C) on a population-weighted basis.

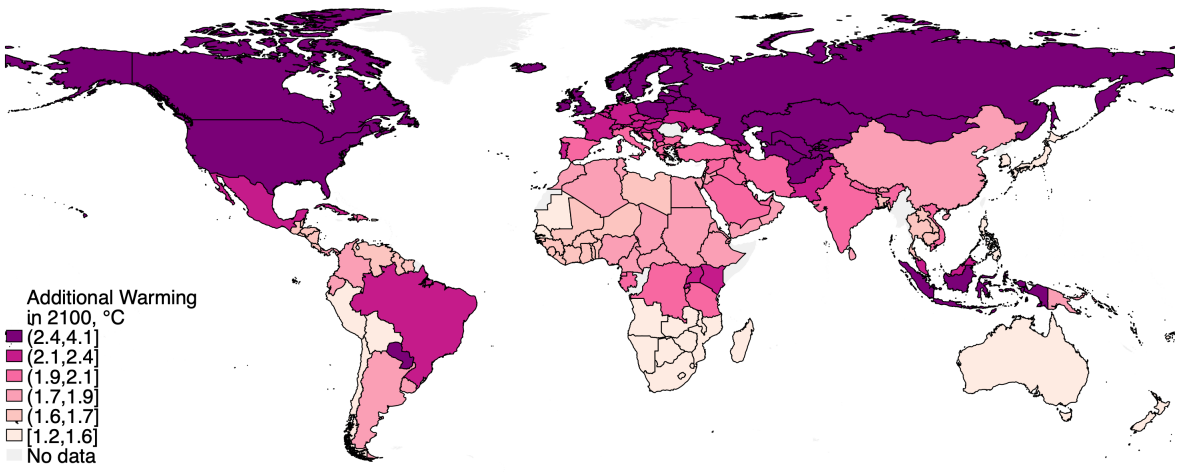



Figure 4: Forecast Warming in 2100 (°C) Under RCP 4.5

[25]The level of warming in Figure 4 is roughly 1 °C less than warming from pre-industrial levels under either RCP as it is measured relative to a 2010-2014 baseline.

**Damages:** Figure 5 maps the warming in Figure 4 into changes in cumulative productivity growth by 2100 using damage values based on estimates from Nath, Ramey, and Klenow (2023). Each country is colored according to how climate change will affect cumulative productivity growth relative to the baseline using the mapping in Equation (16) that simplifies the damage functions in NRK. Countries in the deepest shade of blue experience the largest drags on productivity growth, while those in dark green experience the largest gains from warming. As productivity losses from climate change follow today's temperature gradient, the sign and magnitude of the changes in productivity growth induced by warming vary quite a bit globally; today's warm places are forecast to experience the largest losses while cold ones experience large gains. The unweighted mean (median) change in country-level cumulative productivity growth through 2100 is a loss of 5.5 (10.9) percentage points (p.p.). This decline is substantial – it represents a loss in cumulative growth equivalent to more than five (ten) years of modeled baseline growth over that period. The lost productivity growth is highly diffuse geographically – the standard deviation in losses is 13 percentage points. While warm countries like Indonesia experience losses of 20 p.p. of growth, the coldest countries today, such as Iceland and Mongolia, experience gains of over 50 additional percentage points due to climate change.

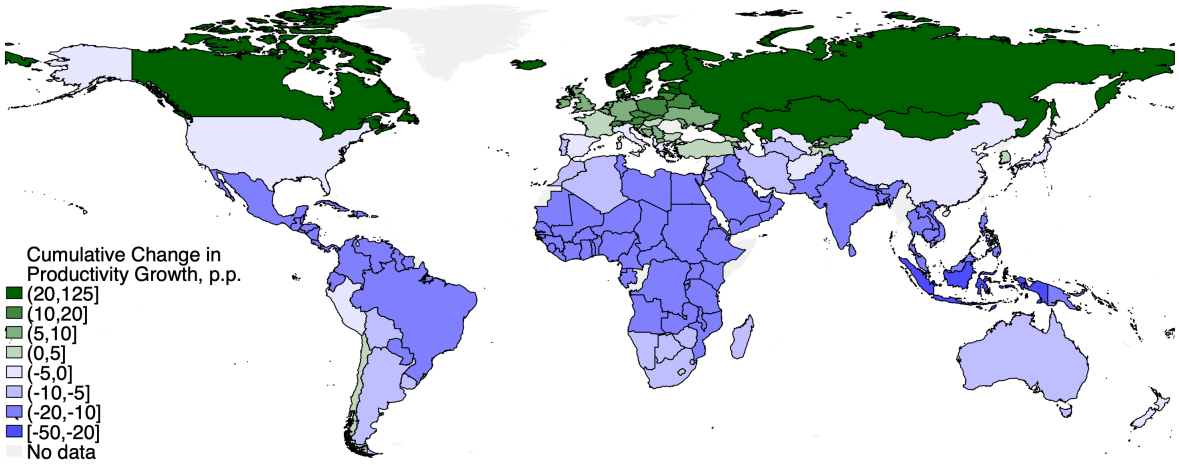



Figure 5: Cumulative Change in Productivity Growth by 2100 (p.p.) Induced by Climate Change

**Population Changes:** Figure 6 shows how the global population distribution changes in the new steady state under RCP4.5 relative to the modeled version of 2010-2014. Countries in darker shades of blue (green) experience larger declines (increases) in their shares of global population, measured in percentage points, under the new climate when compared to the

baseline distribution calibrated to 2010-2014. Unlike in other spatial models (cf. Cruz and Rossi-Hansberg 2023), the gradient of population changes is not monotonically increasing in latitude, despite net northward migration towards regions that gain from climate change.

Six percent of the world population relocates (on net) to different countries after the cumulative effects of warming through 2100 are fully realized. The largest population declines as percentages of initial (steady-state) population levels occur in the United States, Australia, Japan, and Western Europe. This result is not because these locations are the worst hit, but instead because the agents in these locations have sufficient income that more of them prefer to move to locations in Northern Europe or Canada that experience large productivity gains. A non-negligible amount of net emigration occurs in places that are relatively cold initially, as these baseline cold places tend to be near even colder ones that experience even larger gains. The model predicts an especially large decline in the population of the United States (almost 50% of baseline levels, approximately 119 million people) as emigration to Canada is an inexpensive way to trade forecast local damages for the gains north of the border.

Despite the large degree of net emigration from large countries in response to climate change, the amount of emigration from the worst-hit poor countries is limited. Forecast changes in populations in Sub-Saharan Africa, a region where baseline productivities are relatively low and the impacts of climate change are forecast to be very negative, are all lower than 0.1 percent of the global population on net. This occurs because a large fraction of agents who reside in poor countries are unable to borrow or save up to the point of having enough wealth to move. Their only avenue for adaptation is then to increase savings locally, either to smooth income in place or eventually to meet the costs of moving.

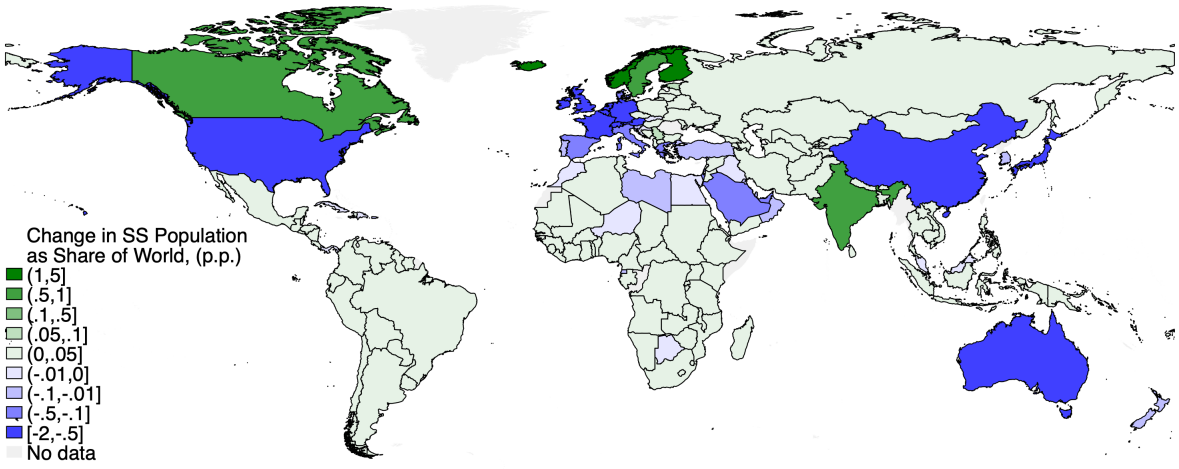



Figure 6: Change in Country-Level Shares of Global Population in New Steady State under RCP 4.5.

## 6.2 Transition Analysis under RCP 4.5

For the transition analysis, I initialize the model in its pre-climate change steady state and feed in a trajectory of warming between 2015 and 2100 based on the projections from the modeling groups participating in CMIP6. After the shock is realized, agents have correct expectations over the trajectory of warming (and in turn, all other endogenous aggregate variables) over the entire transition period. Given a conjectured path for these aggregate variables, I solve agents' value and policy functions backward from the terminal period, so that decisions in each period incorporate their implications for continuation values over the remainder of the transition. I then simulate the distribution of agents forward using these policy functions, updating the implied paths of aggregate variables until the conjectured and realized transition paths converge. The resulting sequence of interest rates, wages, and population distributions each year is the rational expectations equilibrium during the 21$^{st}$ century as the world warms according to RCP 4.5 and approaches the new steady states in Section 6.1. The full algorithm for solving for the transition paths is given in Appendix G and closely resembles standard approaches (e.g., Conesa and Krueger 1999).

**Population Dynamics:** A first output of the transition analysis is the full path of how the global population in the model responds to climate change in each year afer the emissions pathway under RCP 4.5 begins in 2015. Figure 7 shows how the population distribution (green line) and global temperatures (blue line) change each year relative to the calibrated steady state from 2015 through the new steady state after 2100 (shown as a cross-section in Figure 6). The blue line shows the population-weighted change in average annual temperatures under RCP 4.5 based on projections from the CMIP6 project; for years after 2100, I assume temperatures settle to the level reached that year. The green line show how the share of agents in each country changes relative to the baseline steady state, as measured by the sum over all countries of the absolute value of the change in the share of the global population living there. Larger values indicate that countries globally have experienced larger absolute changes in population shares relative to the baseline 2010-2014 period.

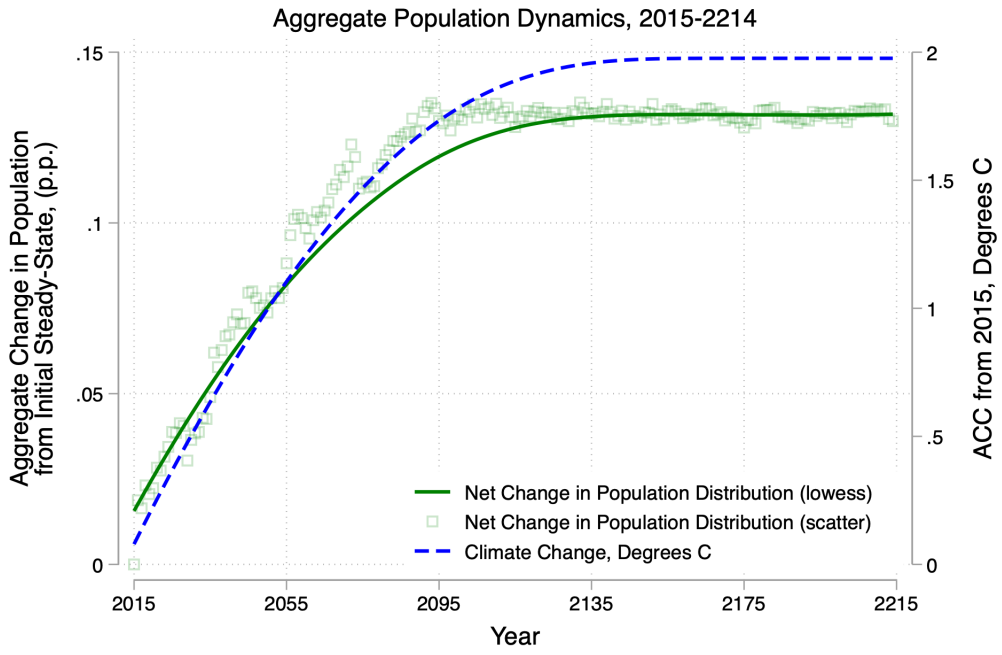



Figure 7: Annual cumulative change in the equilibrium population distribution and GMST (anthropogenic climate change) for the simulated 2015 to 2215 period. The green and blue lines show lowess-smoothed fits of the individual values for changes in population shares and $\Delta$GMST each year. Light green markers denote individual annual outcomes for changes in the simulated population distribution and display some volatility due to simulation noise.

By 2100, climate change induces a 12 percentage point gross change in country-level population shares. This corresponds to 6 percent of the global population changing where they live internationally (on net) in response to climate change. An important aspect of this transition is its speed: the labor side of the model converges to the new steady-state distribution of agents over time quite quickly as opposed to being a delayed response to climate change. This occurs because agents react to both realized changes in local wages due to warming as well as the changes in expectations of future wages they foresee along the transition path. At the global level, adaptation to climate change through migration moves closely with realized warming after a large initial reaction in the years after 2015 when agents learn of the shock.[26] This large initial shift in agents' location choices highlights prior research on the importance of expectations over future climate change in driving anticipatory adaptation (Bilal and Rossi-Hansberg 2023).

**Aggregate Welfare:** Solving for the transition path between steady states allows for an

[26] Aggregate savings adjusts even more quickly than migration as there are no frictions on the global capital market. See Appendix Figure H.1.

apples-to-apples comparison of how climate change affects the welfare of those who experience it. A comparison between outcomes under the initial equilibrium with the steady states of Section 6.1 ignores that wages and interest rates also change along the transition path. If the new steady state only comes about after a substantial amount of expenditure on moving as well as higher savings, welfare may be higher on average (or even globally) in the new steady state once the distribution has settled while at the same timing having caused large declines in consumption during the transition period. To have an accurate accounting of the costs incurred due to climate change, the correct counterfactual needs to compare these costs of adjustment along the transition path to what agents' welfare would have been in 2015 absent RCP 4.5 being realized.

Towards calculating the effect of RCP 4.5 on aggregate welfare, let $\hat{\Omega}_0$ and $\mathbb{E}\left[\hat{V}_0(\hat{x}, \hat{g}_0(\hat{x}, j))\right]$ denote the initial steady state distribution and continuation values in the economy calibrated to the 2010-2014 period. For each agent, given their optimal savings decision $\hat{g}_0(\hat{x}, j)$ conditional on location choice $j$, they go into modeled 2015 expecting to have a continuation value of $\mathbb{E}\left[\hat{V}_0(\hat{x}, \hat{g}_0(\hat{x}, j))\right]$ given they believe the economy remains in steady state. However, the continuation values in 2015 will instead be $\mathbb{E}\left[\hat{V}_{\mathcal{T},1}(\hat{x}, \hat{g}_0(\hat{x}, j))\right]$, the expected value associated with policy $\hat{g}_0(\hat{x}, j)$ given that the realized continuation values will end up corresponding to the first year of the transition path (denoted by subscript $_{\mathcal{T},1}$ here). While agents do not know this when their destinations and conditional savings levels $\hat{g}_0(\hat{x}, j)$ in 2014, once in 2015 rational expectations over climate change leads to knowing the full equilibrium pathway such that their future behavior will induce a value of $\mathbb{E}\left[\hat{V}_{\mathcal{T},1}(\hat{x}, \hat{g}_0(\hat{x}, j))\right]$ in expectation. Summing the changes in realized continuation values over all agents gives a measure of change in social welfare due to climate change driven by RCP 4.5:

$$\frac{\Delta \text{SW}}{\text{SW}} = \sum_{\hat{x}} \sum_{j=1}^{N} \left( \frac{\mathbb{E}\left[\hat{V}_{\mathcal{T},1}(\hat{x}, \hat{g}_0(\hat{x}, j))\right]}{\mathbb{E}\left[\hat{V}_0(\hat{x}, \hat{g}_0(\hat{x}, j))\right]} - 1 \right) \hat{\Omega}_0(\hat{x})\ \pi(\hat{x}, j). \tag{23}$$

The expression in equation (23) expresses the change in aggregate social welfare as the average percentage change in welfare across all agents in the initial steady state of the model. This average of changes under the RCP 4.5 scenario proves to be essentially zero.[27] The small average welfare effect occurs not because effects are small at the individual level, but rather because gains end up offsetting losses. I now turn to describing the heterogeneity that drives this limited change in aggregate outcomes.

**Heterogeneity:** To examine the heterogeneous welfare effects of climate change, I calcu-

[27] The point estimate is $\frac{\Delta \text{SW}}{\text{SW}} = 0.1$, or a 0.1% increase on average.

late the consumption-equivalent of welfare changes for each agent. This measure expresses the change in expected lifetime utility as an equivalent permanent percentage change in consumption relative to the initial steady state. Under logarithmic utility, multiplying consumption in every period and state by $1 + \Delta\text{CE}/100$, holding other components of utility fixed, changes expected lifetime utility by $\log(1 + \Delta\text{CE}/100)/(1 - \beta)$. Equating this expression to the difference in expected lifetime utility between the climate change transition and the initial steady state, and using the approximation $\log(1 + z) \approx z$ for small $z$, gives

$$\Delta\text{CE} \approx 100 \times (1 - \beta) \times \left( \mathbb{E}\left[\hat{V}_{\mathcal{T},1}(\hat{x}, \hat{g}_0(\hat{x}, j))\right] - \mathbb{E}\left[\hat{V}_0(\hat{x}, \hat{g}_0(\hat{x}, j))\right] \right). \tag{24}$$

Here, a negative $\Delta\text{CE}$ indicates the percentage reduction in consumption in every period under the initial steady state that would leave a given agent indifferent between that reduced consumption path and the climate change transition. The expression therefore translates a change in lifetime utility into a permanent consumption equivalent: the percentage loss in lifetime consumption in a world where the climate never changes that leaves agents indifferent between that reduced consumption path and a world under RCP 4.5.

Figure 8 shows the distribution of consumption-equivalent effects across all agents in the initial steady state. As with aggregate welfare, the mean (median) change due to RCP 4.5 is small, with effects equivalent to a 1.4 (2.5) percent increase in lifetime consumption. This masks substantial heterogeneity: welfare changes on a consumption-equivalent basis have a standard deviation of 11.3 percentage points across all agents. Climate change causes the worst-hit agents—agents with zero assets who begin in Indonesia, those in the leftmost bin of Figure 8—to experience losses equal to a 15 percent decline in lifetime consumption. For agents who begin in the lowest decile of asset holdings, RCP 4.5 induces a loss of 8.2 percent in consumption-equivalent terms. In contrast, agents who begin in the highest decile of asset holdings (who hold over 1000 times the average wealth of the bottom decile) gain the most from climate change; they realize increases in welfare equivalent to a 13.6 percent increase in lifetime consumption.

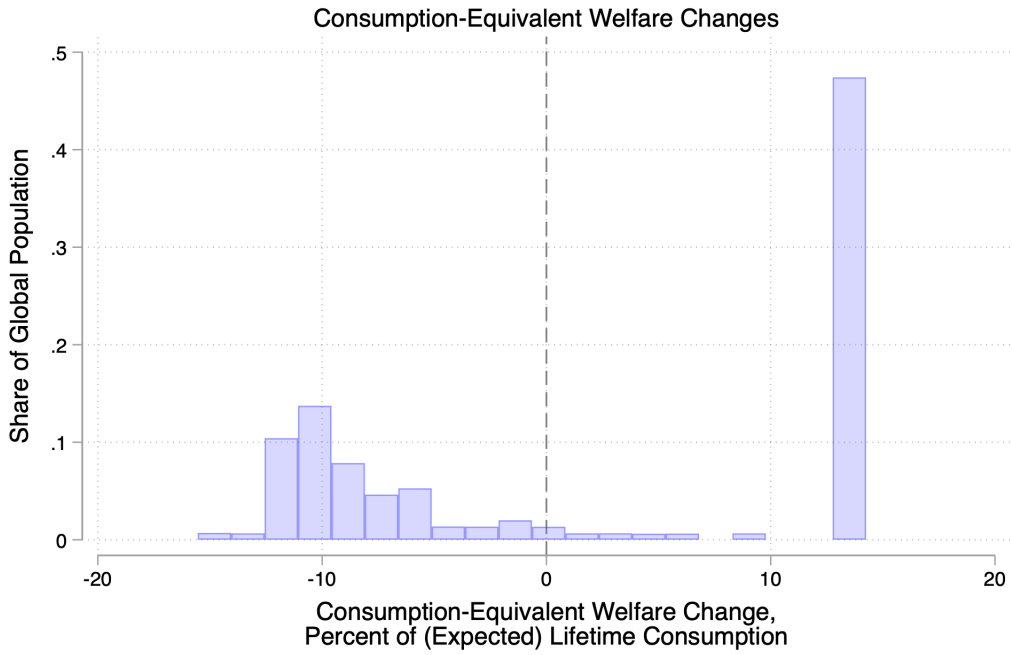



Figure 8: Distribution of Agent-Level Losses under RCP 4.5, Expressed in Consumption Equivalents.

Figure 9 illustrates how the gains and losses from climate change are distributed across agents with different initial standings in the economy. The plot shows changes in consumption-equivalent welfare gains by deciles of agents' (expected) continuation utility given their last decision and state variables under the initial steady state prior to entering the first period of the transition path. The world is roughly split between winners and losers, and the losers are concentrated among the initially worst-off individuals. The bottom 50 percent of the welfare distribution today experience losses of about 3 percent on average, while the best-off decile today experiences gains from climate change of almost 4 percent on average. Appendix Figure H.2 shows that this pattern of inequality also holds if we instead bin agents by their initial wealth holdings (as opposed to initial welfare levels that account for the effects of financial wealth and location). On the whole, results from the transition path add to the growing body of research that suggests the welfare effects of climate change will be highly unequal by geography as well as across income groups within a given country.




Figure 9: Average Welfare Change by Initial Welfare Level Decile

## 6.3 Alternative Counterfactuals

I consider two alternative counterfactuals for how future climate change may evolve. I present only the long-run (i.e., steady-state) forecasts here, as they are more informative than the transition paths in these cases. The first alternative re-examines what the steady state would look like under a more pessimistic scenario for warming induced by the RCP 8.5 emissions pathway. The second examines how the steady state changes if, in addition to the productivity effects of warming under RCP 4.5, migration costs are concurrently doubled globally.

**RCP 8.5:** Figure 10 displays the analogue of Figure 6 for the changes in steady-state populations after the world warms by 4.2 °C on average under the RCP 8.5 scenario. Appendix Figures I.1 and I.2 show the associated changes in country-level average temperatures and cumulative productivity growth. The extreme productivity changes induced by this pessimistic scenario for emissions cause massive shifts in the global population distribution; 21 percent of the global population migrates internationally relative to the baseline steady state. However, for the most part, the forecast changes under RCP 8.5 are exaggerated versions of those in the main model. The lion's share of migration in Figure 10 consists of bilateral flows between rich countries and northward moves toward the Scandinavian countries that

experience large additional growth gains (over 70 additional percentage points!).

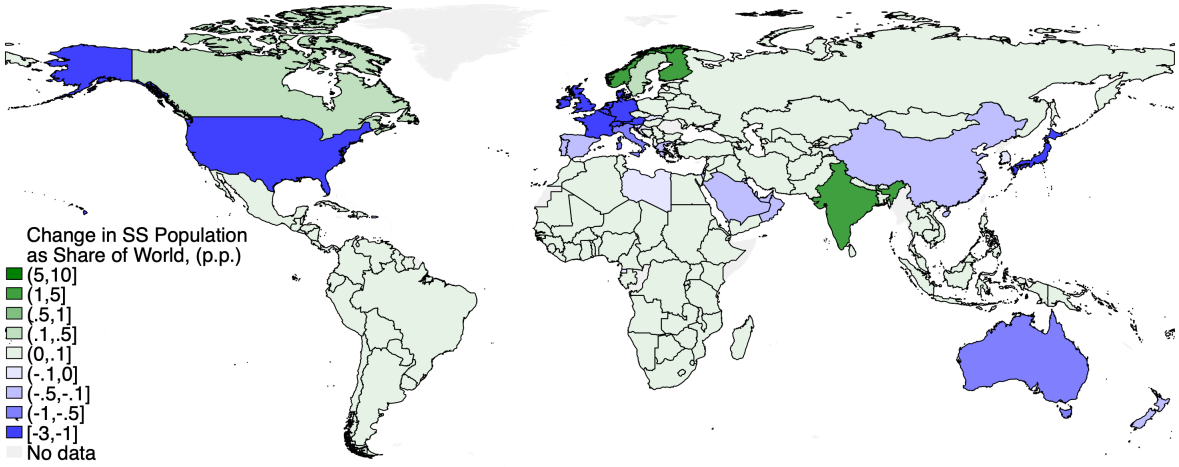



Figure 10: Change in Country-Level Shares of Global Population in New Steady State, RCP 8.5

**Doubling of Migration Costs:** This scenario considers a world in which, in addition to the warming under RCP 4.5, bilateral migration costs $m(i, j)$ are doubled between all country pairs. This reflects the common belief that international migration policies may become less accommodating in the 21$^{st}$ century, especially toward economic migration induced by climate change (Vince 2022). The resulting change in the distribution of the global population is shown in Figure 11. Doubling migration costs makes agents substantially less sensitive to taste shocks driving migration choices that do not follow the gradient of productivity. As such, almost all modeled agents either remain stuck in areas far from Northern Europe where wages are low or end up in Northern Europe and mix only within the Scandinavian countries that receive large productivity gains from warming. The magnitude of the changes under this scenario renders the forecast somewhat difficult to interpret; I instead view these results as indicative of the extreme importance of considering financial migration costs directly and taking care to ensure they are calibrated correctly.

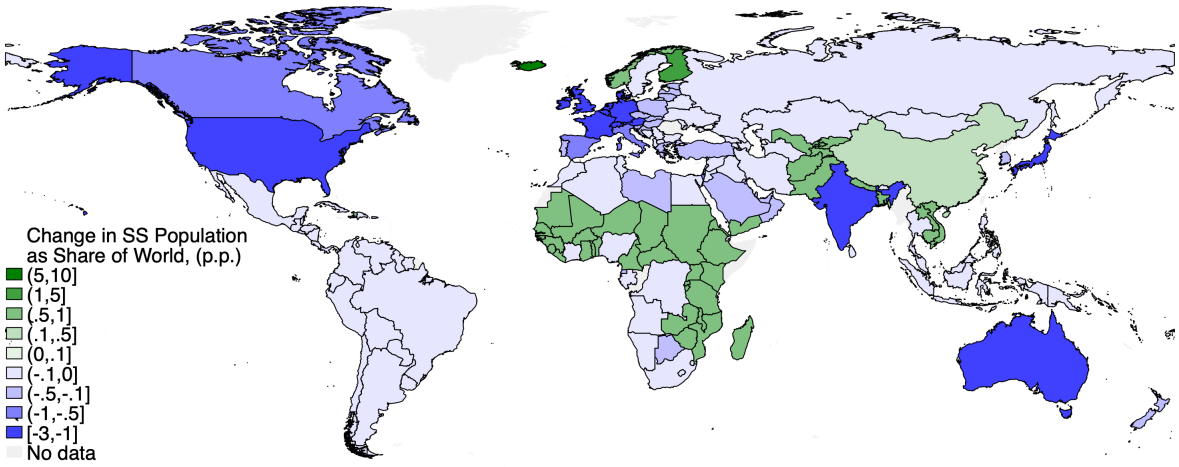



Figure 11: Change in Country-Level Shares of Global Population in New Steady State, Double Migration Costs

# 7 Conclusion

This paper presents a global model of population dynamics under climate change in the 21st century. The model nests a neoclassical consumption-savings decision within a benchmark dynamic discrete-choice spatial framework in order to examine how local consumption patterns and migration will collectively shape adaptation to climate change. The primary goal of the model is to take seriously the notion that impediments to migration manifest not only as lost utility but also as pecuniary costs. Requiring migrants to pay a cost reflects the reality that many people who might otherwise gain from migration from a utility standpoint are unable to do so because of limited financial resources (Bazzi 2017; Cai 2020).

The broad prediction of the simple and calibrated models is that the value of international migration as adaptation will be determined on an individual basis based on ability to pay. Modeling migration as having pecuniary costs—beyond pure disutility—leads to sharp nonlinearities in the modeled migration policy functions, a feature absent from existing state-of-the-art regional macroeconomic models of climate change. These nonlinearities capture the observed pattern that, in poor countries, emigration is increasing in income (Cai et al. 2016; Bazzi 2017; Barbosa-Alves and Braulio 2023). If future barriers to migration manifest as monetary costs as well as disutility, the model strongly suggests we should expect to see stark differences between the effects of climate change on agents who can afford to migrate and those who cannot.

My main experiment in this paper examines how the global gradient of productivity changes induced by climate change under RCP 4.5 shifts the global population during the

21st century and over a longer-run horizon. The model projects that 6 percent of the global population relocates to different countries when the cumulative effects of warming through 2100 are realized. Despite common narratives suggesting that emigration will be concentrated in the Global South (Vince 2022), migration is instead concentrated among rich countries and is used for adaptation almost exclusively by agents who hold higher places in the global wealth distribution. This occurs because the barriers to international migration are large enough that many poor agents find it too expensive to adapt through the migration channel and instead are forced to rely solely on self-insurance. The inequitable access to migration as adaptation causes the effects of climate change at the individual level to be highly unequal across poor and rich agents. While the aggregate effects of climate change on welfare under RCP 4.5 are small, poor agents experience welfare losses equivalent to an 8.2 percent decline in lifetime consumption, while the richest agents experience welfare gains equivalent to a 13 percent increase.

This first step in bridging the gap between the neoclassical and spatial approaches (Krusell and Smith 2022; Cruz and Rossi-Hansberg 2023) suggests that agents' behavior when they have access to adaptation across space and time may not resemble a convex combination of the outcomes that would occur when each channel is accessible in isolation. My combination of the two approaches is not without drawbacks; the model here omits a number of attractive features unique to the above models, including making climate change and energy use endogenous, allowing for incompleteness in global capital markets, and accounting for the twin effects of crowding and agglomeration when considering how agents select locations. Other simplifying assumptions lead the model to overpredict the appeal of small countries in the Global North, which causes the model to understate the number of agents who are negatively affected by climate change. Nevertheless, the preliminary work here suggests that future work on spatial models of climate change that characterizes how income heterogeneity shapes individual-level adaptation decisions promises new insights into how these decisions will affect global aggregates.

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# Appendix

## Contents


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# A Proof of Proposition 1

*Proof.*

**Part 1 of Proposition 1:** An increase in $m$ raises $\pi(i,i)$.

Consider the expression for $\pi(i,i)$:

$$\pi(i,i) = \frac{1}{\sum_{j=1}^{N} e^{W(i,j)-W(i,i)}}. \tag{A.1}$$

Substituting for each $W(i,j)$ and noting the optimal savings policy $S^*(i,j)$ is unaffected by $U$ as since the utility migration cost is additive and does not depend on savings, interior solutions for optimal savings solves agents' Euler equations:

$$\begin{aligned} \frac{1-\beta}{y-m(i,j)-S(i,j)} &= \frac{\beta R}{w_2(j)+RS(i,j)} \\ S^*(i,j) &= \max\left\{\frac{\beta R\left(y-m(i,j)\right)-(1-\beta)w_2(j)}{R}, \frac{-w_2(j)}{R}\right\}, \end{aligned} \tag{A.2}$$

where $R = (1+r)$ is the gross return to savings and the max operator ensures the agent does not violate the natural borrowing constraint in (1). For all feasible destinations $j$, the conditional value is then

$$W(i,j) = \ln\left(\frac{\left((1-\beta)(y-m(i,j)) + \frac{(1-\beta)w_2(j)}{R}\right)^{1-\beta}}{\left(\beta R(y-m(i,j)) + \beta w_2(j)\right)^{-\beta}}\right) - U\mathbb{I}\{j \neq i\}.$$

Since the utility cost is zero for staying, the differences $W(i,j) - W(i,i)$ become

$$W(i,j) - W(i,i) = \ln\left(\frac{R(y-m(i,j)) + w_2(j)}{Ry + w_2(i)}\right) - U\mathbb{I}\{j \neq i\}. \tag{A.3}$$

It's useful at this point to define

$$\eta(i,j) \triangleq e^{-U\mathbb{I}\{j\neq i\}} > 0.$$

to write fewer indicator functions. Substituting the $W(i,j)-W(i,i)$ terms in Equation (A.1) then lets us write migration probabilities as

$$\pi(i,j) = \max\left\{ \frac{\eta(i,j)\left[R(y-m(i,j))+w_2(j)\right]}{\sum_{k=1}^{N}\eta(i,k)\max\left\{R(y-m(i,k))+w_2(k),0\right\}}, 0 \right\}. \tag{A.4}$$

Differentiating (A.4) in emigration costs from island $i$ gives

$$d\pi(i,i) = \frac{\left[-\pi(i,i)\frac{\partial}{\partial m}\left\{\sum_{j\in\mathcal{J}(i)}\eta(i,j)\left[R(y-m(i,j))+w_2(j)\right]\right\}\right]}{\sum_{j\in\mathcal{J}(i)}\eta(i,j)\left[R(y-m(i,j))+w_2(j)\right]}dm. \tag{A.5}$$

where $\eta(i,i)=1$. The summations are restricted to the set of period-2 destinations that, for an agent starting on island $i$ in period 1, allow for positive consumption in the first period:

$$\mathcal{J}(i) = \left\{ j \in \{1,\ldots,N\} \mid y + w_2(j)(1+r)^{-1} - m(i,j) > 0 \right\}.$$

Suppose

$$\frac{\partial m(i,j)}{\partial m} > 0, \qquad \forall i \neq j,$$

and

$$\frac{\partial m(i,i)}{\partial m} = 0.$$

Rearranging and noting that the denominator is positive implies

$$\begin{aligned}\frac{d\pi(i,i)}{dm} &\propto -\pi(i,i)\frac{\partial}{\partial m}\left\{\sum_{j\in\mathcal{J}(i)}\eta(i,j)\left[R(y-m(i,j))+w_2(j)\right]\right\} \\ &= \pi(i,i)R\sum_{j\in\mathcal{J}(i)}\eta(i,j)\frac{\partial m(i,j)}{\partial m} \geq 0.\end{aligned}$$

which is the stated result. The inequality is strict whenever the set $\mathcal{J}(i)$ contains at least one feasible destination other than $i$.

**Part 2 of Proposition 1:** A decline in $A_2(j)$ lowers $\pi(i,j)$.

When $\pi(i,j) = 0$ initially because location $j$ is infeasible, it is trivial that $\pi(i,j)$ cannot be any lower. For the case where $\pi(i,j) > 0$ initially, taking its derivative in $A_2(j)$ gives

$$\frac{d\pi(i,j)}{dA_2(j)} = \frac{\eta(i,j)\tilde{w} - \pi(i,j)\frac{\partial}{\partial A_2(j)}\left\{\sum_{k\in\mathcal{J}(i)}\eta(i,k)\left[R(y-m(i,k))+w_2(k)\right]\right\}}{\sum\limits_{k\in\mathcal{J}(i)}\eta(i,k)\left[R(y-m(i,k))+w_2(k)\right]}.$$

where

$$\tilde{w} = (1-\alpha)\left(\frac{\alpha}{r}\right)^{\frac{\alpha}{1-\alpha}}$$

is the wage per effective worker. As the denominator is positive, applying the derivative and rearranging gives the desired result

$$\frac{d\pi(i,j)}{dA_2(j)} \propto \eta(i,j)\tilde{w}(1-\pi(i,j)) > 0, \tag{A.6}$$

as $\eta(i,j) > 0$, $\tilde{w} > 0$, and $\pi(i,j) < 1$.

□

# B Proof of Proposition 2

*Proof.* To prove Proposition 2, let

$$\tilde{w}'(r) = -\alpha\left(\frac{\alpha}{r}\right)^{\frac{1}{1-\alpha}} < 0$$

be the change of effective wages with the interest rate offered to firms. Shorthand

$$Q(i) \triangleq \sum_{j\in\mathcal{J}(i)}\eta(i,j)$$

and define the utility-cost-weighted mean of second-period productivity and travel costs across each island $j$ that receives a nonzero share of agents from island $i$ in the second period:

$$\bar{A}_U(i) = \frac{1}{Q(i)}\sum_{j\in\mathcal{J}(i)}\eta(i,j)A_2(j), \qquad \overline{m_U(i)} = \frac{1}{Q(i)}\sum_{j\in\mathcal{J}(i)}\eta(i,j)m(i,j),$$

where the summations are again restricted to the set of feasible destinations from origin $i$:

$$\mathcal{J}(i) = \Big\{ j \in \{0, \ldots, N\} \mid y + w_2(j)(1+r)^{-1} - m(i,j) > 0 \Big\}.$$

Define

$$D \triangleq R(y - \overline{m_U(i)}) + \widetilde{w}(r)\bar{A}_U(i).$$

Since $\eta(i,i) = 1$, the share of stayers is

$$\pi(i,i) = \frac{Ry + \widetilde{w}(r)A_2(i)}{Q(i)D}.$$

Holding the feasible set fixed locally, differentiate $\pi(i,i)$ in $r$ to give

$$\begin{aligned}
\frac{d\pi(i,i)}{dr} &= \frac{[y + \widetilde{w}'(r)A_2(i)]D - [Ry + \widetilde{w}(r)A_2(i)][y - \overline{m_U(i)} + \widetilde{w}'(r)\bar{A}_U(i)]}{Q(i)D^2} \\
&= \underbrace{\frac{\widetilde{w}(r) - R\widetilde{w}'(r)}{Q(i)D^2}}_{>0} \left[ y(\bar{A}_U(i) - A_2(i)) + A_2(i)\overline{m_U(i)} \right].
\end{aligned}$$

Since $\widetilde{w}(r) > 0$ and $\widetilde{w}'(r) < 0$, the sign of the derivative is determined by the bracketed expression. If $A_2(i) < \bar{A}_U(i)$, its first term is strictly positive and its second term is nonnegative, so

$$\frac{d\pi(i,i)}{dr} > 0.$$

When $A_2(i) > \bar{A}_U(i)$, the first term is negative and the second is nonnegative, making the effect ambiguous. Specifically,

$$\frac{d\pi(i,i)}{dr} > 0 \quad \Longleftrightarrow \quad \overline{m_U(i)} > y\left(1 - \frac{\bar{A}_U(i)}{A_2(i)}\right). \tag{B.1}$$

Thus an increase in $r$ increases $\pi(i,i)$ in locations where their productivity is below the utility-cost-weighted average productivity in potential destinations $\mathcal{J}(i)$, while its effect in locations more productive than this weighted average depends on migration costs. Larger weighted average migration costs $\overline{m_U(i)}$ in (B.1) make increasing interest rates more likely to increase $\pi(i,i)$ even in more-productive locations.

□

# C Proof an Equilibrium Exists for the Two-Period Model

*Proof.* To prove at least one equilibrium exists for this economy requires finding an interest rate at which equation (5) holds when both aggregate demand and supply of capital are generated by the optimal policies of agents and firms at interest rate $r$. A sufficient condition for existence of an equilibrium is that the excess demand function for capital (that is, aggregate demand less supply as a function of $r$) is continuous and takes values above and below zero. I show that (1) aggregate capital supply and demand are continuous on $r \in (0, \infty)$ and (2) excess demand is positive for sufficiently small $r$ and negative for sufficiently large $r$.

**Proving (1):** Letting $R = 1 + r$ again denote the gross interest rate, wages in the second period are

$$w_2(i) = \tilde{w}(r)A(i),$$

where as above

$$\tilde{w}(r) = (1 - \alpha)\left(\frac{\alpha}{r}\right)^{\frac{\alpha}{1-\alpha}}.$$

We can rewrite the migration probabilities (A.4) as functions of $r$:

$$\pi(j,i) = \max\left\{\frac{\eta(j,i)\left[R(y - m(j,i)) + \tilde{w}(r)A(i)\right]}{\sum_{k \in \mathcal{J}(j)} \eta(j,k)\left[R(y - m(j,k)) + \tilde{w}(r)A(k)\right]}, 0\right\}. \tag{C.1}$$

Each term in the numerator and denominator is continuous in $r$ on $(0, \infty)$. As $m(j,j) = 0$ and $\eta(j,j) = 1$, the denominator is bounded below by

$$Ry + \tilde{w}(r)A(j) > 0.$$

Thus migration probabilities are continuous for all origin and destination pairs. This remains true even when a destination enters or exits the set of feasible destinations $\mathcal{J}(j)$, as the probabilities need not be differentiable at these thresholds but remain continuous.

As each island initially contains a unit measure of agents, second-period populations satisfy

$$N_2(i) = \sum_{j=1}^{N} \pi(j,i), \qquad \sum_{i=1}^{N} N_2(i) = \sum_{i=1}^{N} 1 = N.$$

By (6), aggregate capital demand in the second period is then

$$K^D(r) = \left(\frac{\alpha}{r}\right)^{\frac{1}{1-\alpha}} \sum_{i=1}^{N} A(i)N_2(i)$$
$$= \left(\frac{\alpha}{r}\right)^{\frac{1}{1-\alpha}} \sum_{i=1}^{N} A(i) \sum_{j=1}^{N} \pi(j,i). \tag{C.2}$$

As migration probabilities are continuous on $r \in (0, \infty)$, aggregate capital demand in (C.2) is also continuous on this interval.

Turning to supply, for each feasible destination from island $i$, optimal savings are unchanged by the additive utility migration cost:

$$S(i,j) = \frac{\beta R(y - m(i,j)) - (1-\beta)\tilde{w}(r)A(j)}{R}.$$

Aggregate supply is then

$$K^S(r) = \sum_{i=1}^{N} \sum_{j=1}^{N} \pi(i,j) \left[\frac{\beta R(y - m(i,j)) - (1-\beta)\tilde{w}(r)A(j)}{R}\right].$$

Because infeasible destinations have $\pi(i,j) = 0$, we need not worry about the potential for the numerator inside the brackets to violate the constraint that consumption be positive. By inspection, the bracketed expression is continuous and finite at every $r > 0$, including at feasibility thresholds. Its product with $\pi(i,j)$ is therefore continuous. As capital supply is a finite sum of these products, $K^S(r)$ is continuous on $(0, \infty)$.

Both aggregate capital supply and demand are then continuous on this interval, which ensures their difference (excess demand) is continuous.

**Proving (2):** Consider first the limit as $r \to \infty$. Since

$$\lim_{r \to \infty} \frac{\tilde{w}(r)}{R} = 0,$$

dividing the numerator and denominator of equation (C.1) by $R$ gives

$$\lim_{r \to \infty} \pi(i,j) = \frac{\eta(i,j) \max\{y - m(i,j), 0\}}{\sum_{k=1}^{N} \eta(i,k) \max\{y - m(i,k), 0\}}.$$

The denominator is strictly positive because $m(i,i) = 0$, $\eta(i,i) = 1$, and $y > 0$. Substituting these limiting probabilities as well as the formula for optimal savings (A.2) into the formula

for aggregate capital supply gives

$$\lim_{r\to\infty} K^S(r) = \beta \sum_{i=1}^{N} \frac{\sum_{j=1}^{N} \eta(i,j)(\max\{y - m(i,j), 0\})^2}{\sum_{k=1}^{N} \eta(i,k) \max\{y - m(i,k), 0\}} > 0.$$

The last inequality is strict because, for each origin $i$, staying contributes $y^2 > 0$ to the numerator as $m(i,i) = 0$ and $\eta(i,i) = 1$. As the total population is mass $N$ and second-period productivity levels are assumed bounded below positive infinity, it follows that

$$\lim_{r\to\infty} K^D(r) = 0.$$

Next consider the limit as $r \to 0^+$. Capital demand in (C.2) diverges:

$$\lim_{r\to 0^+} K^D(r) = \infty,$$

as long as the aggregate effective labor supply is positive (i.e., there is some island $i$ for which $\theta(i) < \infty$).

To establish the corresponding limit for capital supply, rewrite aggregate savings as

$$K^S(r) = \beta \sum_{i=1}^{N}\sum_{j=1}^{N} \pi(i,j)(y - m(i,j)) - (1-\beta)\frac{\tilde{w}(r)}{R} \sum_{i=1}^{N}\sum_{j=1}^{N} \pi(i,j)A(j).$$

The first term is bounded because migration costs and probabilities are finite and the probabilities sum to one for each origin. The utility cost changes the distribution of the $\pi(i,j)$ terms but not this property. As the sum multiplying $\tilde{w}(r)/R$ in the second term is strictly positive by assumption, and $\tilde{w}(r)/R \to \infty$, this establishes

$$\lim_{r\to 0^+} K^S(r) = -\infty.$$

The excess demand function

$$Z(r) = K^D(r) - K^S(r)$$

therefore satisfies

$$\lim_{r\to 0^+} Z(r) > \infty, \qquad \lim_{r\to\infty} Z(r) = -\lim_{r\to\infty} K^S(r) < 0.$$

There consequently exist two finite positive interest rates at which excess demand has opposite signs. As $Z(r)$ is continuous on $(0, \infty)$, the Intermediate Value Theorem implies that it equals zero at least once between these rates. At this interest rate, aggregate capital supply equals demand, and the optimal policies of agents and firms constitute an equilibrium in Definition 1.

□

# D Ambiguity in the General Case

*Proof.*

**Part 1:** Write $m'(i,j) = \partial m(i,j)/\partial m$. From Appendix C, the share of stayers is

$$\pi(i,i) = \frac{Ry + A_2(i)\tilde{w}(r)}{\sum_{j=1}^{N} \eta(i,j)\left[R(y - m(i,j)) + A_2(j)\tilde{w}(r)\right]},$$

where terms associated with infeasible destinations are understood to have zero probability. Along a differentiable equilibrium and away from feasibility thresholds, differentiating with respect to $m$ gives

$$\frac{d\pi(i,i)}{dm} = \frac{\frac{dr}{dm}\left(y + A_2(i)\tilde{w}'(r)\right) - \pi(i,i)\sum_{j\in\mathcal{J}(i)} \eta(i,j)\left[\frac{dr}{dm}\left(y - m(i,j) + A_2(j)\tilde{w}'(r)\right) - Rm'(i,j)\right]}{\sum_{j\in\mathcal{J}(i)} \eta(i,j)\left[R(y - m(i,j)) + A_2(j)\tilde{w}(r)\right]}.$$

The denominator is positive, so $d\pi(i,i)/dm$ is signed by

$$\frac{dr}{dm}\left[y + A_2(i)\tilde{w}'(r) - \pi(i,i)\sum_{j\in\mathcal{J}(i)} \eta(i,j)\left(y - m(i,j) + A_2(j)\tilde{w}'(r)\right)\right] + \pi(i,i)R\sum_{j\in\mathcal{J}(i)} \eta(i,j)m'(i,j).$$

Assuming $m'(i,j) \geq 0$ for every destination, the final term is nonnegative. This is the direct effect of increasing migration costs while holding the interest rate fixed which led to the results in Proposition 1. The first term captures the general-equilibrium effect operating through the equilibrium interest rate. Because the bracketed term and $dr/dm$ need not have fixed signs, this effect can offset the direct effect. The effect of a change in migration costs on the share of stayers is therefore ambiguous in general equilibrium.

**Part 2:** Differentiating the share of migrants from origin $j$ choosing destination $i$ with respect to $A_2(i)$ gives

$$\frac{d\pi(j,i)}{dA_2(i)} = \frac{\begin{array}{r}\eta(j,i)\dfrac{dr}{dA_2(i)}\left(y - m(j,i) + A_2(i)\tilde{w}'(r)\right) + \eta(j,i)\tilde{w}(r) \\ - \pi(j,i)\left\{\dfrac{dr}{dA_2(i)}\displaystyle\sum_{k\in\mathcal{J}(j)}\eta(j,k)\left[y - m(j,k) + A_2(k)\tilde{w}'(r)\right] + \eta(j,i)\tilde{w}(r)\right\}\end{array}}{\displaystyle\sum_{k\in\mathcal{J}(j)}\eta(j,k)\left[R(y - m(j,k)) + A_2(k)\tilde{w}(r)\right]}.$$

The denominator is again positive. Consequently, $d\pi(j,i)/dA_2(i)$ has the same sign as

$$\frac{dr}{dA_2(i)}\left[\eta(j,i)\left(y - m(j,i) + A_2(i)\tilde{w}'(r)\right) - \pi(j,i)\sum_{k\in\mathcal{J}(j)}\eta(j,k)\left(y - m(j,k) + A_2(k)\tilde{w}'(r)\right)\right] + \eta(j,i)\tilde{w}(r)(1 - \pi(j,i)).$$

The final term—the direct effect of increasing productivity on island $i$ while holding the interest rate fixed—is weakly positive. The remaining term captures the effect operating through the equilibrium interest rate and again has no determinate sign.

□

# E Proof of Ambiguity Under a Unique Equilibrium

*Proof.* Following Appendix C, let

$$Z(r,\Theta) = K^D(r,\Theta) - K^S(r,\Theta) = 0$$

characterize the equilibrium condition for zero excess capital demand, where $\Theta$ collects the exogenous parameters including $m(\cdot,\cdot)$, $A(i)$, and $U$. Suppose that capital demand and supply are continuously differentiable and satisfy

$$\frac{\partial K^D}{\partial r} < 0, \qquad \frac{\partial K^S}{\partial r} > 0.$$

almost everywhere. These conditions imply that the equilibrium is unique and that $\partial Z/\partial r < 0$.

Holding all exogenous parameters other than the set of migration costs $m$ fixed, differ-

entiate the equilibrium condition:

$$0 = \left(\frac{\partial K^D}{\partial m} - \frac{\partial K^S}{\partial m}\right) dm + \left(\frac{\partial K^D}{\partial r} - \frac{\partial K^S}{\partial r}\right) dr.$$

Consequently,

$$\frac{dr}{dm} = \frac{\dfrac{\partial K^D}{\partial m} - \dfrac{\partial K^S}{\partial m}}{\dfrac{\partial K^S}{\partial r} - \dfrac{\partial K^D}{\partial r}}.$$

Given the assumption of monotone aggregate capital supply and demand, the denominator is positive. The sign of $dr/dm$ then equals the sign of the change in excess demand induced by $dm$ holding the interest rate fixed.

Recall capital demand and supply are

$$K^D(r, \Theta) = \left(\frac{\alpha}{r}\right)^{\frac{\alpha}{1-\alpha}} \sum_{j \in \mathcal{J}(i)} \sum_{i=1}^{N} A_2(i)\pi(j,i),$$
$$K^S(r, \Theta) = \sum_{j \in \mathcal{J}(i)} \sum_{i=1}^{N} \pi(j,i) S(j,i),$$

where, for each origin $j$, the migration probabilities are now proportional to

$$\pi(j,i) \propto \eta(j,i) \left[R(y - m(j,i)) + A_2(i)\tilde{w}(r)\right]$$

over feasible destinations. Holding $r$ fixed, we have

$$\frac{\partial K^D}{\partial m} = \left(\frac{\alpha}{r}\right)^{\frac{\alpha}{1-\alpha}} \sum_{j \in \mathcal{J}(i)} \sum_{i=1}^{N} A_2(i) \frac{\partial \pi(j,i)}{\partial m},$$
$$\frac{\partial K^S}{\partial m} = \sum_{j \in \mathcal{J}(i)} \sum_{i=1}^{N} S(j,i) \frac{\partial \pi(j,i)}{\partial m} + \sum_{j \in \mathcal{J}(i)} \sum_{i=1}^{N} \pi(j,i) \frac{\partial S(j,i)}{\partial m}.$$

For the interior savings policies,

$$S(j,i) = \beta(y - m(j,i)) - \frac{(1-\beta)A_2(i)\tilde{w}(r)}{R},$$

so the additive utility migration cost does not enter the conditional savings policy and

$$\frac{\partial S(j,i)}{\partial m} = -\beta m'(j,i).$$

It follows that

$$\frac{\partial Z}{\partial m} = \sum_{j \in \mathcal{J}(i)} \sum_{i=1}^{N} \left[ \left( \frac{\alpha}{r} \right)^{\frac{\alpha}{1-\alpha}} A_2(i) - S(j,i) \right] \frac{\partial \pi(j,i)}{\partial m} + \beta \sum_{j \in \mathcal{J}(i)} \sum_{i=1}^{N} \pi(j,i) m'(j,i).$$

The second term is nonnegative when migration costs are non-decreasing. This captures the direct effect of higher pecuniary migration costs on savings policies conditional on agents' destination choices; higher $m(j,i)$ leaves less income available to save for the $j \to i$ migrants and in turn lowers aggregate capital supply. The first term captures the changes in savings and capital demand that occur because migration probabilities also shift. Increasing $m$ changes which agents go where, affecting aggregate capital demand as well as the distribution of optimal savings policies. Hence $\partial Z/\partial m$, and therefore $dr/dm$, remain ambiguous even when a unique equilibrium is assumed.

□

# F Inner Problem Solution Methods

Step two in the solution method of Section 4 requires solving stationary version of agents' dynamic programming problem in (13). The results in the main paper solve the agent's problem using value function iteration (VFI) over a 500 node asset grid. This discretizes the agent's savings problem; $\hat{a}'$ must be an element of this grid as opposed to a continuous choice. Discretizing the problem may lead to numerical issues in terms of how closely the numerical solution matches the theoretical (continuous) solution. The state-of-the-art computational solution would be to instead use the discrete choice endogenous grid method (DCEGM) developed by Iskhakov et al. (2017). The DCEGM deals with the potential nonconvexities while preserving speed by exploiting the analytical invertability of the Euler equation when utility over consumption is logarithmic. This avoids having to call a numerical maximizer at every state space node during VFI. I lay out the algorithm here; while it does converge in a reasonable amount of time for a fixed set of model parameters (order 1 hour), achieving the speed needed for the method of simulated moments step is a work in progress . [28]

[28] The algorithm is fully described here outside of the step in line 18. The upper envelope step of Iskhakov et al. (2017) rules out dominated gridpoints, points in the endogenous grid where the associated conditional value functions $V^n(\omega, j, k, s)$ fail to be monotonic in wealth. See Algorithms 1 and 3 in Iskhakov et al. (2017) for details.

**Algorithm 1** The Modified Iskhakov et al. (2017) DCEGM Algorithm

**Input:** Deep parameters, interest rate $r^i$ and associated wages $\{w_j\}_{j=1}^N$ implied by (9), and migration costs $m(j,k)$.

Fix the exogenous asset grid $A = \{a_p\}_{p=1}^P$, with $a_1 = 0$ and $a_p < a_{p+1}$.

Initialize endogenous wealth grids of length $|A|+1$ for each origin-destination-labor state pair, $M^0(j,k,s)$. Initialize associated conditional consumption and value functions of length $|A|+1$ and $|A|$ denoted as $c^0(\omega,j,k,s)$ and $V^0(\omega,j,k,s)$ which map endogenous wealth $\omega$ into consumption levels and continuation values.

Set $n = 0$ and $e > \varepsilon$.

**while** $e > 10^{-6}$ **do**

  **for** $s \in \{\ell, h\}$ and $k \in \{1,\dots,N\}$ **do**

    **for** $p = 1,\dots,P$ **do** ▷ Loop over savings choices

      **for** $s' \in \{\ell, h\}$ **do**

        Solve for next-period wealth conditional on realization of $s$:

$$\omega'_p = (1+r)a_p + w_j s'.$$

        Interpolate $c^n(\omega'_p, k, d', s')$ and $V^n(\omega'_p, k, d', s')$ on $M^n(k, d', s')$ for each $d' \in \{1,\dots,N\}$.

        Calculate implied CCPs by evaluating conditional value functions at realized state

$$\pi^n(\omega'_p, k, d', s') = \frac{\exp\big(V^n(\omega'_p, k, d', s')/\nu\big)}{\sum_{l=1}^{N} \exp\big(V^n(\omega'_p, k, l, s')/\nu\big)}.$$

      **end for**

      Invert the Euler to solve for the implied level of optimal consumption today

$$c_p^{n+1}(j,k,s) = \left[\frac{\beta(1+r)}{\gamma} \sum_{s'\in\{\ell,h\}} \Pi_{ss'} \sum_{d'=1}^{N} \frac{\pi^n(\omega'_p, k, d', s')}{c^n(\omega'_p, k, d', s')}\right]^{-1}.$$

      **for** $j \in \{1,\dots,N\}$ **do**

        Recover the endogenous wealth point $w_p^{n+1}(j,k,s)$

$$\omega_p^{n+1}(j,k,s) = c_p^{n+1}(j,k,s) + m(j,k) + \gamma a_p - w_k s.$$

        Update conditional value functions

$$V_p^{n+1}(j,k,s) = \ln\big(c_p^{n+1}(j,k,s)\big) + \beta \sum_{s'\in\{\ell,h\}} \Pi_{ss'} \nu \log\left[\sum_{d'=1}^{N} \exp\left(\frac{V^n(\omega'_p, k, d', s')}{\nu}\right)\right].$$

      **end for**

    **end for**

  **end for**

  Apply the upper-envelope step of Iskhakov et al. (2017) to rule out dominated grid points.

  Evaluate convergence on the common wealth grid:

$$e = \max_{\substack{j,k\in\{1,\dots,N\} \\ s\in\{\ell,h\},\ \omega\in M^{n+1}(j,k,s)}} \left|V^{n+1}(\omega,j,k,s) - V^n(\omega,j,k,s)\right|.$$

  Set $n \leftarrow n+1$.

**end while**

**Output:** Converged conditional consumption and value functions $c^n(\omega,j,k,s)$ and $V^n(\omega,j,k,s)$, defined on the endogenous wealth grids $M^n(j,k,s)$

# G Solving for Transition Paths

This appendix provides the details of how to solve for the value and policy functions along a transition path for the model in Section 3 where local fundamental productivity levels change over time. On the transition path, the agent's recursive problem becomes

$$\hat{V}_t\Big(\hat{a}_t, s_t, i_t, \{\lambda_t(j)\}_{j=1}^N\Big) = \max_j \left\{ \max_{\hat{a}_{t+1}} \left\{ u(\hat{c}_t) \;+\; \lambda_t(j) \;+\; \beta\, \mathbb{E}\Big[\hat{V}_{t+1}\Big(\hat{a}_{t+1}, s_{t+1}, j_{t+1}, \{\lambda_{t+1}(k)\}_{k=1}^N\Big)\Big] \right\} \right\} \tag{G.1}$$

$$\text{s.t.} \qquad (1+r_t)\hat{a}_t + \hat{w}_{i,t} s_t = \gamma \hat{a}_{t+1} + \hat{c}_t + \hat{m}(i,j)$$

The key difference along the transition path from the stationary equilibrium is the value functions retain their time indices as-defined in Equation (12). This stems from the fact that the economy is no longer on a balanced growth path; relative wages and interest rates change over time due to the non-stationarity of productivity growth across locations. The objects $\mathbb{E}\Big[\hat{V}_{t+1}\Big(\hat{x}_t, d_t\Big)\Big]$ now solve the functional equation:

$$\mathbb{E}\Big[\hat{V}_{t+1}\Big(\hat{x}_t, \hat{d}_t\Big)\Big] = \sum_{s' \in \ell, h} \Pi_{ss'} \ln\left( \sum_{k_{t+1}} \exp\left( \max_{\hat{a}_{t+2}} \left\{ u_{t+1}\Big(\hat{x}_{t+1}, d_{t+1}\Big) \;+\; \beta\, \mathbb{E}\Big[\hat{V}_{t+2}\Big(x_{t+1}, \hat{d}_{t+1}\Big)\Big] \right\} \right) \right) \tag{G.2}$$

which like in the non-stationary definition in equation (13) of the main text has time indices on all arguments. The algorithm proceeds as follows:

1. Guess a time $T$ when the economy has settled into a new stationary equilibrium after the productivity shocks from climate change have been realized.

2. Solve for the initial SCRE, denoted with subscript zero, comprised of $\hat{\Omega}_0(\hat{x}), \hat{V}_0, r_0, \{\hat{w}_0\}, g_0, \pi_0$ under fundamental productivity levels $\bar{A}_j$.

3. Solve for the new SCRE $\hat{\Omega}_\infty(\hat{x}), \mathbb{E}[\hat{V}_\infty], r_\infty, \{\hat{w}_\infty\}, g_\infty, \pi_\infty$ that is induced by the fundamental productivity levels under fully-realized climate change, $\tilde{A}_j$

4. Assume that at time $T$, the economy along the transition path reaches the new SCRE such that for all times $t \geq T$ we have

$$\mathbb{E}[\hat{V}_{t \geq T}(\hat{x}, \hat{d})] = \mathbb{E}[\hat{V}_\infty(\hat{x}, \hat{d})]$$

and all other equilibrium objects $\pi, \hat{w}, r, g(\hat{x}, j')$ are stationary.

5. Guess a sequence of labor $\{\{N_{i,t}^D\}_{i=1}^N\}_1^{T-1}$ and capital $\{K_t^D\}_1^{T-1}$ demand.

6. Use reverse induction to solve for the series $\{\mathbb{E}[\hat{V}_t]\}_{t=1}^{T-1}$ induced by the guess of aggregate variables along with the sequence of changes in fundamental productivity levels along the transition path

7. Simulate a large number of agents along the transition path and solve for the resulting values of labor supplied $\{\{N_{i,t}\}_{i=1}^S\}_1^{T-1}$ and $\{K_t^S\}_1^{T-1}$

8. Adjust guesses for $\{\{N_{i,t}^D\}_{i=1}^N\}_1^{T-1}$ and $\{K_t^D\}_1^{T-1}$ until supply and demand coincide

9. Check if the distribution of assets and agents $\hat{\Omega}_T(\hat{x})$ induced by the sequence of decision rules is equal to the time-invariant value $\hat{\Omega}_\infty(\hat{x})$ within tolerance

10. If it is, algorithm has converged, if not, increase $T$ and go to step 4.

In practice, I initially set $T = 400$, and it proves to be enough time for convergence.

# H Additional Figures

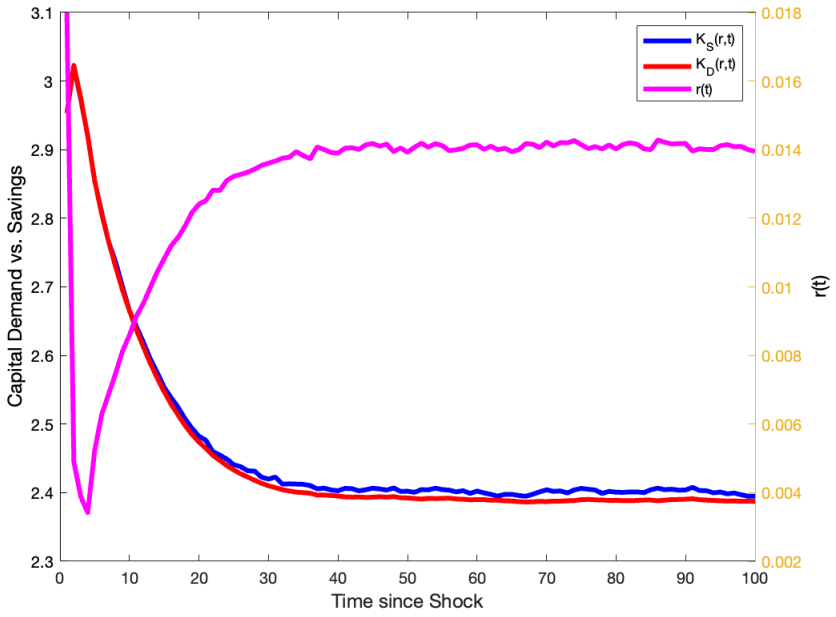



Figure H.1: Convergence of Economic Aggregates Along Transition Path




Figure H.2: Average Consumption-Equivalent Loss by Initial Wealth Level Decile, Percent CE

# I Steady-State Warming and Productivity Changes under RCP 8.5

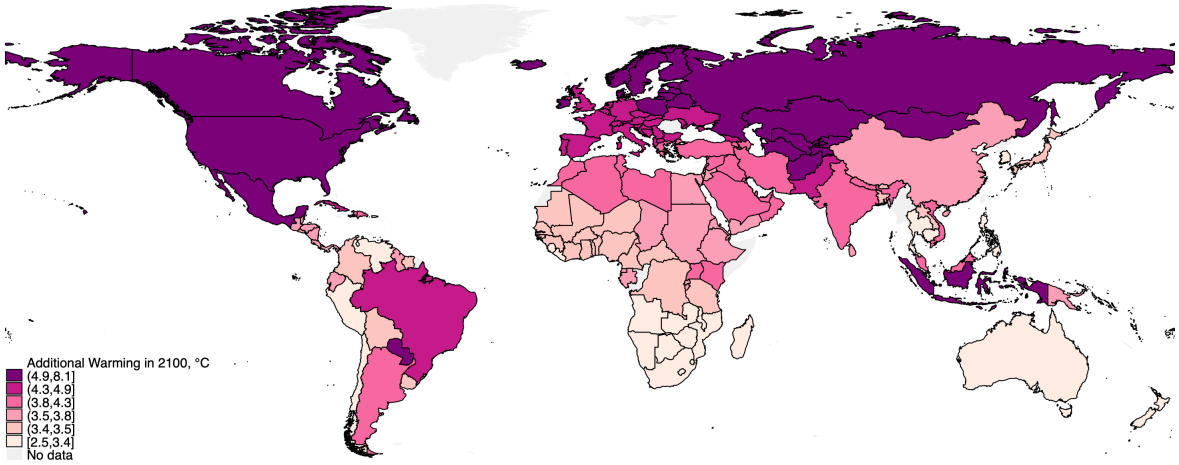



Figure I.1: Forecast Warming in 2100 (°C) Under RCP 8.5

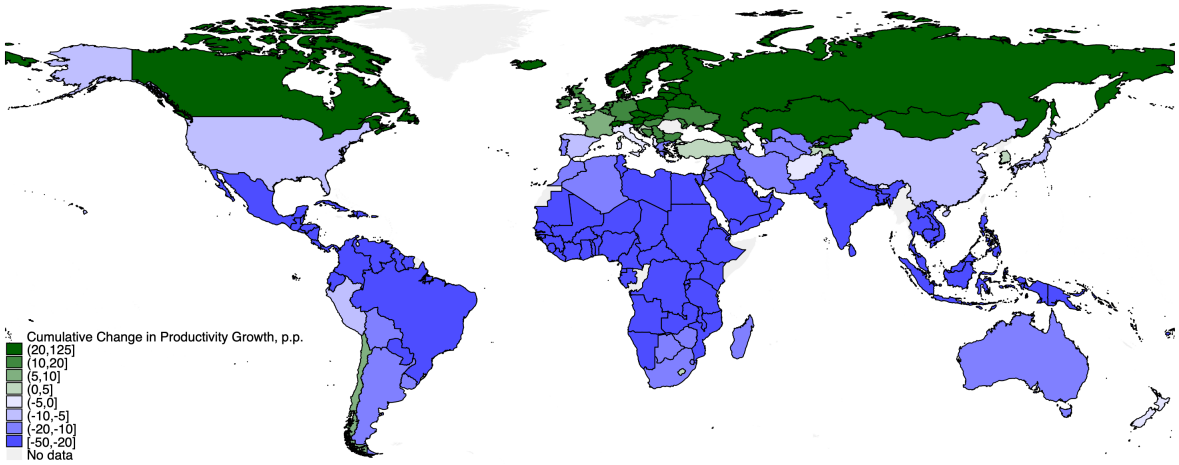



Figure I.2: Cumulative Change in Productivity Growth by 2100 (p.p.) Induced by Climate Change under RCP 8.5